

5GHz Active probe

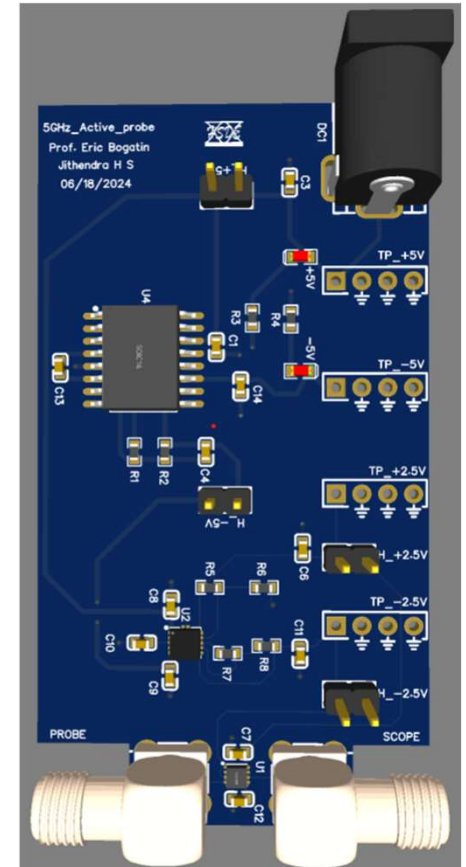
Guided by: Prof. Eric Bogatin

Presented by: Jithendra H S

06/12/2024

Introduction:

- Eval board that can support 5GHz bandwidth.
- Method to characterize probe and active devices (Opamp).
- Support measurement up to signal voltage varying between +1.5V and -1.5V.
- Implement board with best practices by taking care of SI issues.
- Make use of EasyEDA tool(Free/online tool).
- Low-cost probe.



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Mini projects

Tasks	Status	Date	Remarks
Working principles of Passive and Active probes	completed		None
Simulation with spice	completed		Did simulate for LT1054 and opa855, forgot to take screenshots
Component selection and board review	completed	06/18/24	Mentioned in next slides
Measure TDR for Wildriver strip and micro strip transmission lines	completed	06/18/24	
Measure gain bandwidth, insertion and return loss for bandpass filters and attenuators using Nano VNA.	completed	06/19/24	Measured and plotted using ADS
Design DC blocking for RF opamp circuits	completed	06/25/24	Preferred using high capacitive value(20uF) so that it can be reused for low pole frequency ckts
Measure input impedance and output impedance of RF opamps	completed	07/10/24	Measured and plotted using ADS
Measure gain bandwidth of few RF opamps using VNA(measure S21)	completed	07/10/24	Measured and plotted using ADS

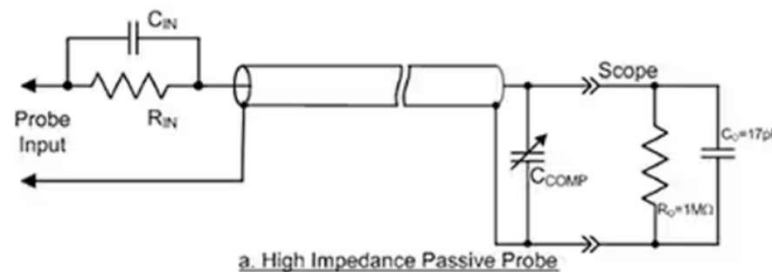
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Tasks	Status	Date	Remarks
Design breakout boards with LT1054 and TPS7A39 for testing	completed	06/29/24	
Measure the 5GHz probe input impedance, output impedance and gain bandwidth.	Completed/needs remeasure	07/10/24	Measured and plotted using ADS
Measure power rail noise(inverter oscillator effect)	completed	07/17/2024	
Breakout board characterization	Completed	10/08/2024	
Characterize the DC blocking capacitor	Completed	10/08/2024	
Design breakout board to sniff USB, HDMI and Ethernet	Ongoing		
Characterize attenuators with and without DC block.	completed	07/17/2024	
Hack to know the capacitance and inductance effect on impedance	completed	07/17/2024	
Documentation			

Introduction

Passive probes:



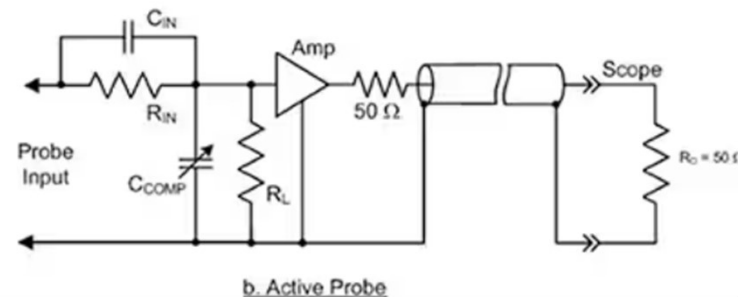
- The most common type of scope probes used, and have bandwidth up to 500MHz.
- High input capacitance(around 10pF) -> DUT loading increases with increase in frequency.

$$I_c = V/X_c, \text{ where } X_c \text{ is capacitive reactance.}$$

Source: <https://www.digikey.com/en/articles/understanding-selecting-using-active-oscilloscope-probes>

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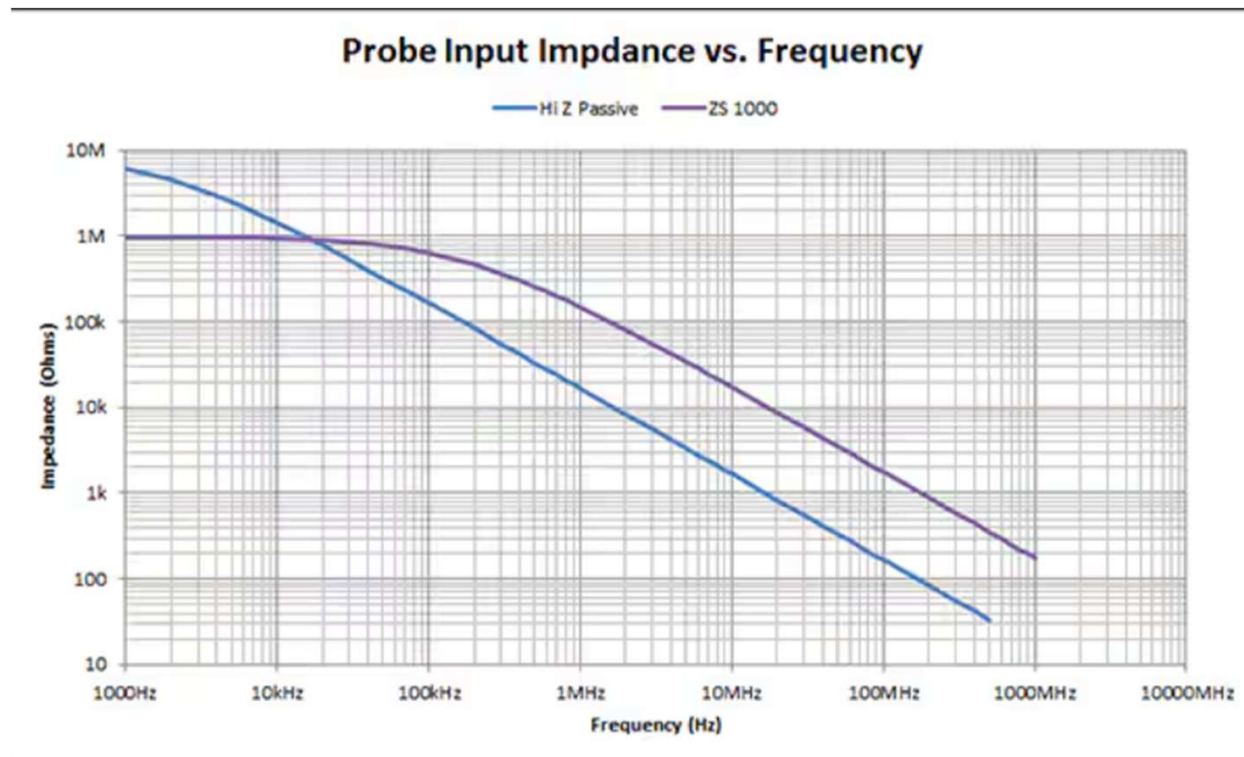
Active probes:



- Are scope probes used to measure signal bandwidth above 500MHz.
- Amplifier buffer allows the cable to be terminated into characteristic impedance isolating the capacitive loading.
- Low input capacitance(around 1pF) due to other hardware component allow this cable to measure higher bandwidth with much loading.
- Need external power source.

Source: <https://www.digikey.com/en/articles/understanding-selecting-using-active-oscilloscope-probes>

Frequency response

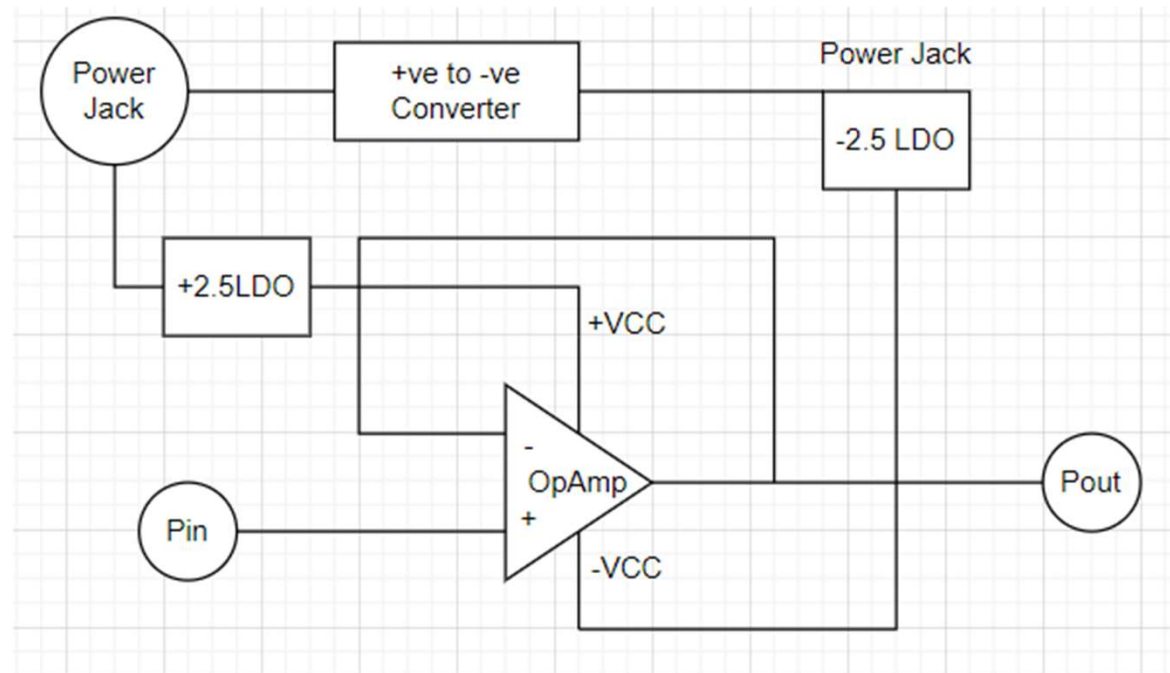


Plan of record

- 5V power supply.
- Choose fast switching OpAmp as buffer.
- Select suitable charge pump(Switched capacitor voltage converter) to convert 5V to -5V need for OpAmp power supply.
- LDO to regulate voltage supply to +2.5V and -2.5V.
- SMA connectors(50ohm impedance) for probe connection.
- Design transmission line length between SMA connector and opAmp non-inverting pin to reduce reflections.
- Follow best practices like:- test points, led indicators and labelling.

Napkin Sketch

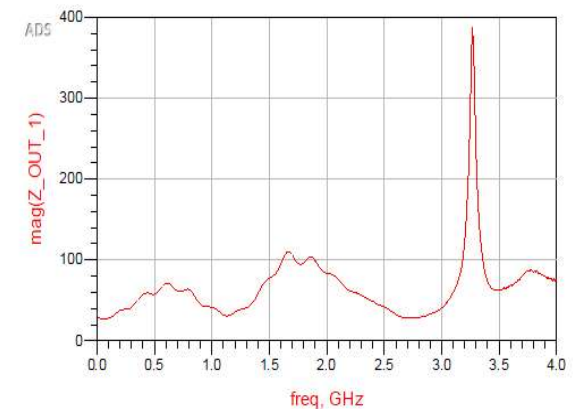
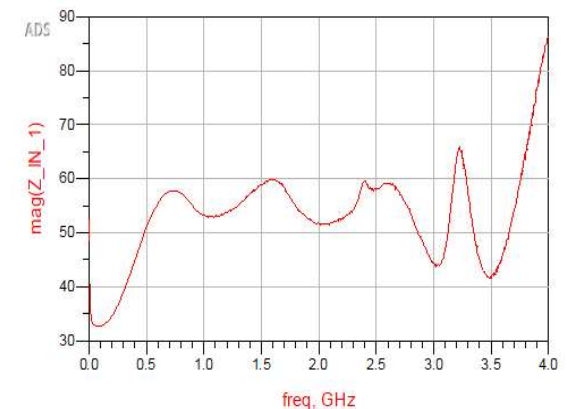
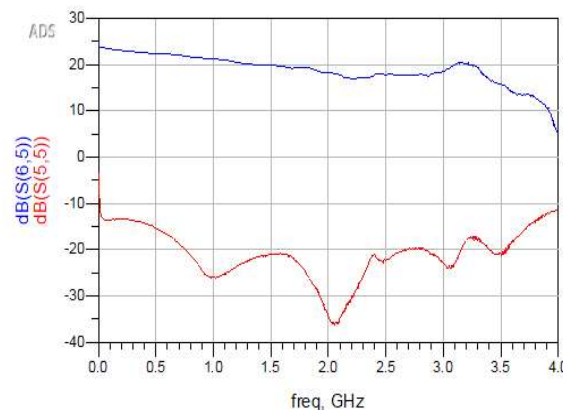
- OpAmp: OPA858/OPA855.
- DC converter: ICL7660/LT1054
- LDO: TPS784-Q1 and TPS723-Q1
- SMA connector
- LEDs
- Decoupling capacitors



S-Parameter analysis of RF amplifiers

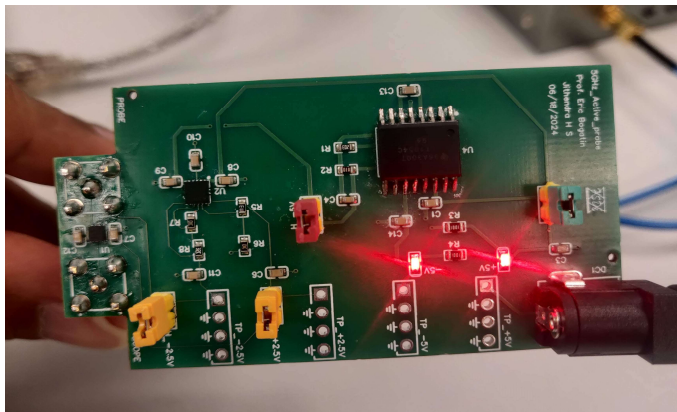


- RF amplifier with 20dB gain at 5-3500MHz
- Used 40dB attenuator to safeguard VNA port and set the port 2 attenuator factor to 40dB
- Observation from the graphs:
 1. Insertion loss vary by 7dB over the operating frequency range 5-3500MHz
 2. Input and output impedance of opamp is around 50ohm but not exact 50ohm, which result in return loss as shown



S-parameter analysis of Active probe

- Tested the basic functionality of designed board using oscilloscope
- Measured S-parameters of the probe, unfortunately our probe can measure small signal bandwidth up to 2GHz without distorting it, but not close to the expected 5GHz.



7.5 Electrical Characteristics

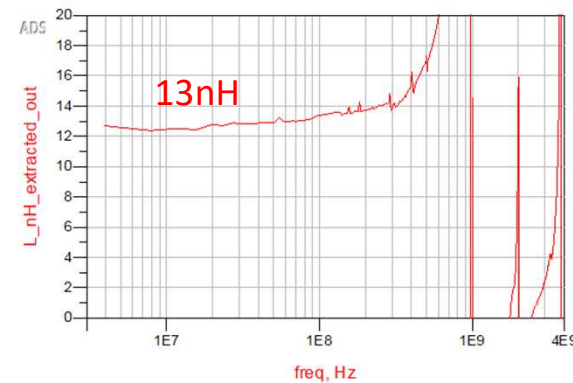
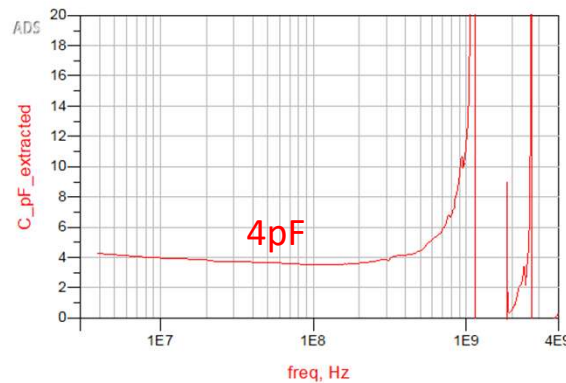
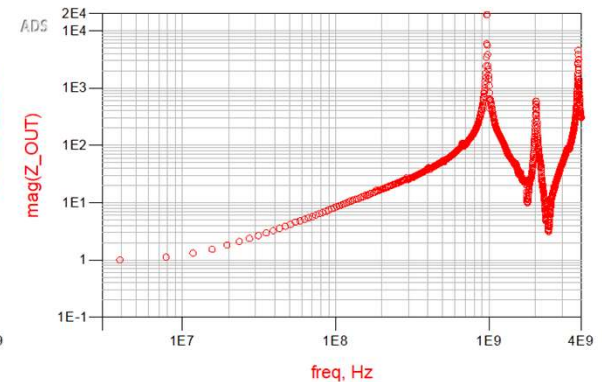
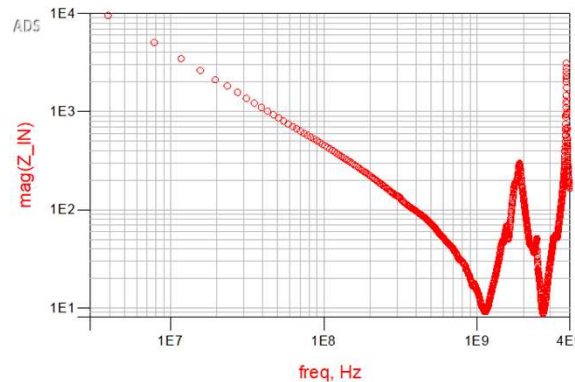
$V_{S+} = 5\text{ V}$, $V_{S-} = 0\text{ V}$, $G = 7\text{ V/V}$, $R_F = 453\ \Omega$, input common-mode biased at midsupply, $R_L = 200\ \Omega$, output load is referenced to midsupply, and $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	TEST LEVEL ⁽¹⁾
AC PERFORMANCE							
SSBW	Small-signal bandwidth	$V_{OUT} = 100\text{ mV}_{PP}$		1.2		GHz	C
LSBW	Large-signal bandwidth	$V_{OUT} = 2\text{ V}_{PP}$		600		MHz	C
GBWP	Gain-bandwidth product			5.5		GHz	C

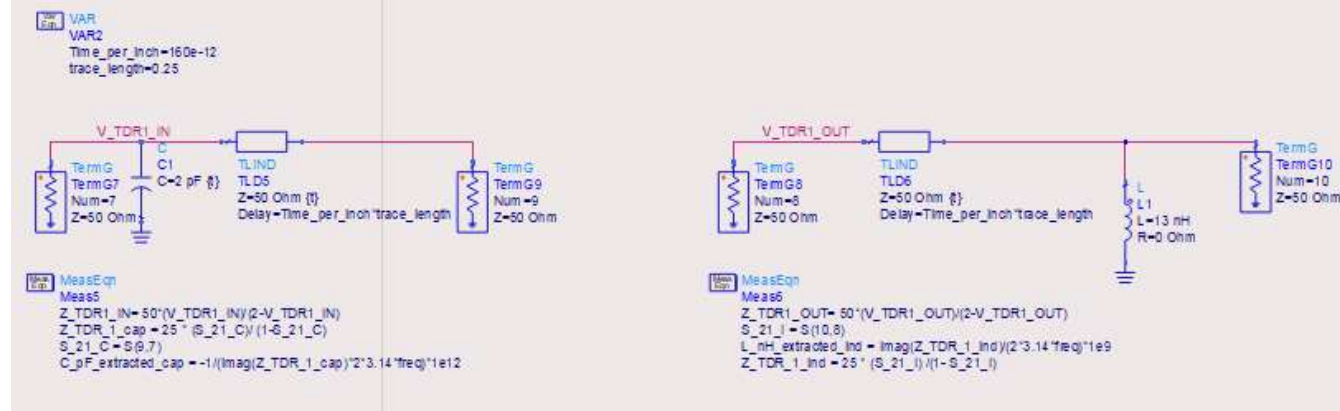
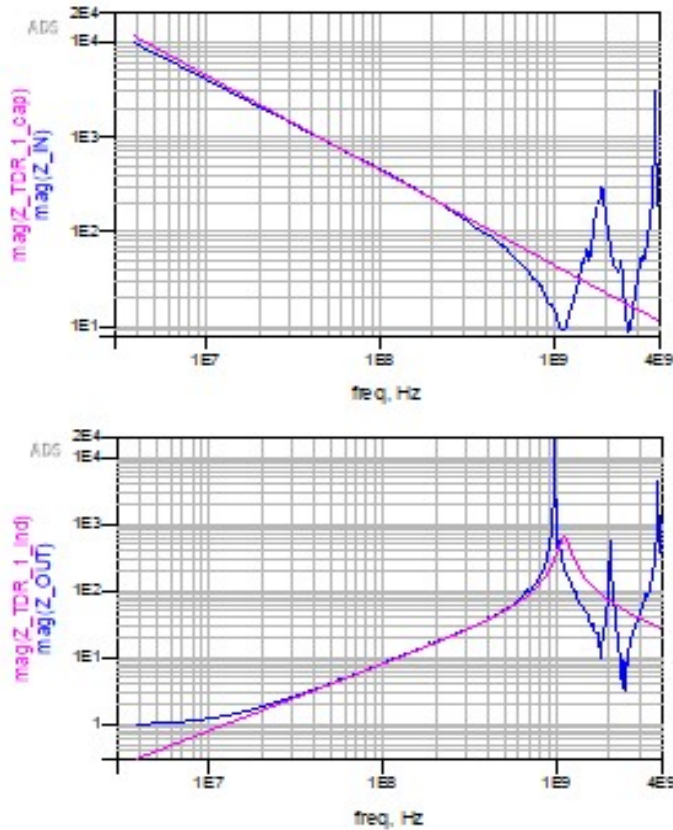


Input and Output Impedance

- Input impedance behavior matches capacitive impedance behavior at lower frequencies ($C = 1/(2 * \pi * f * Z_C)$), and it is around magnitude of 4pF.
- Output impedance behaviour matches inductive impedance at lower frequencies ($L = Z_L/(2 * \pi * f)$), and it is around magnitude of 13nH.



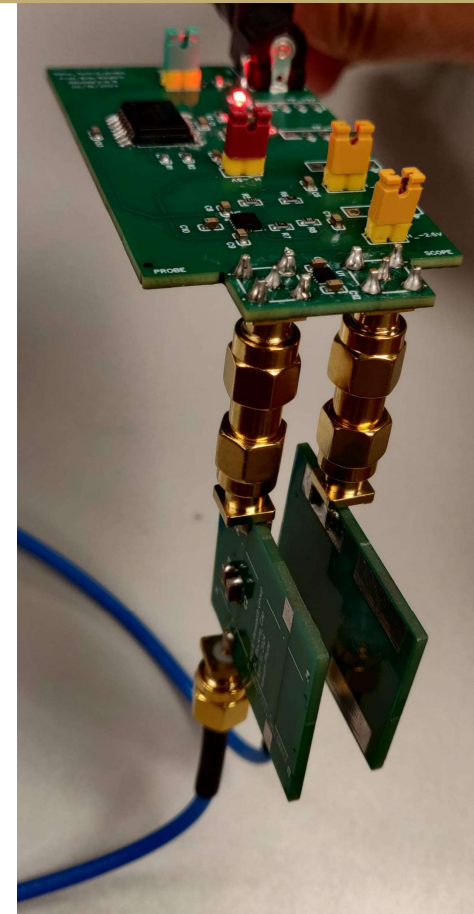
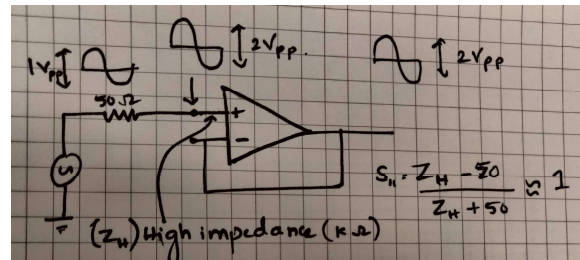
Hacking the input and output impedances



Simulated result shows the capacitance could be **2pF** and inductance of about **13nH** and they are pretty good match to the S-parameter findings from earlier slide

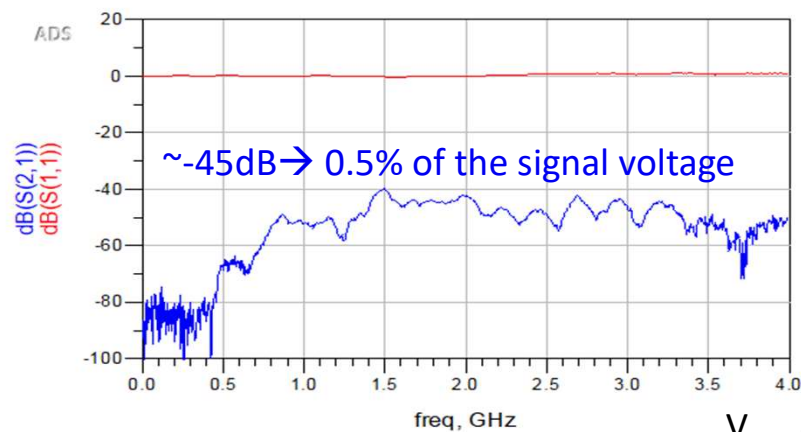
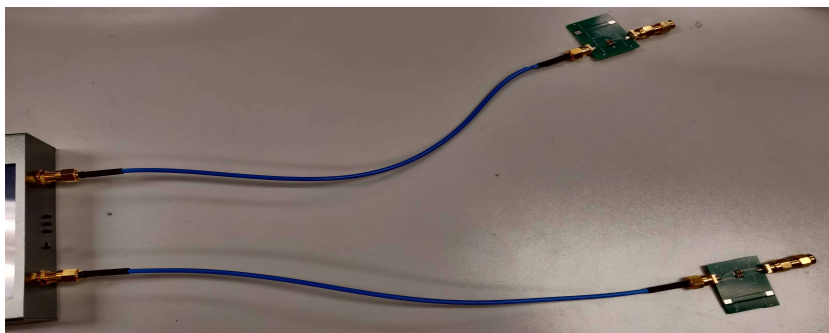
Justification for 6dB output

- Opamp usually have high input impedance, so the reflection coefficient ~ 1 and the input terminal sees twice the amplitude signal (6dB). The voltage follower circuit with gain 0 result in 6dB signal.

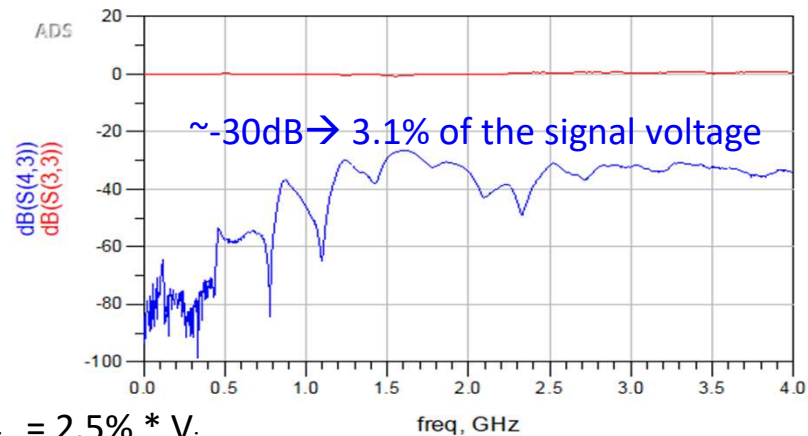
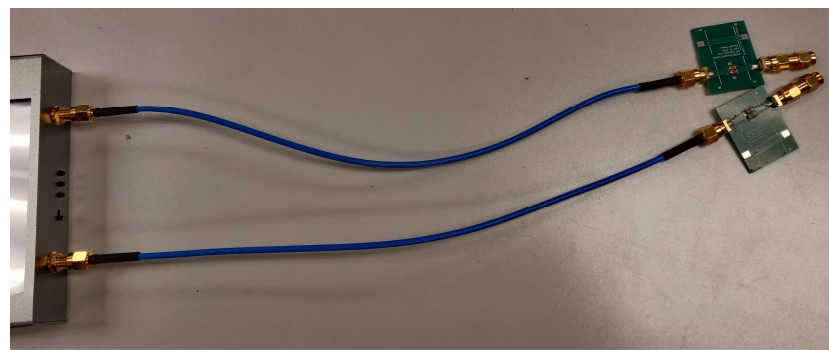


Coupling:

Cable ends are far apart



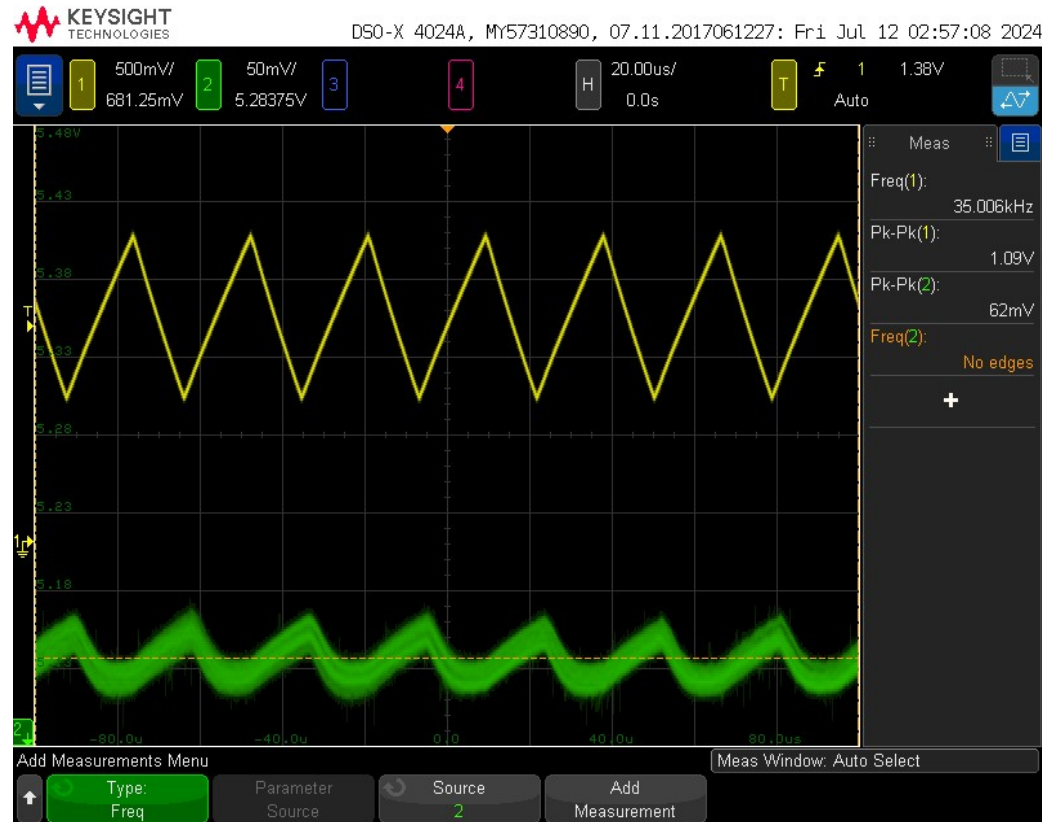
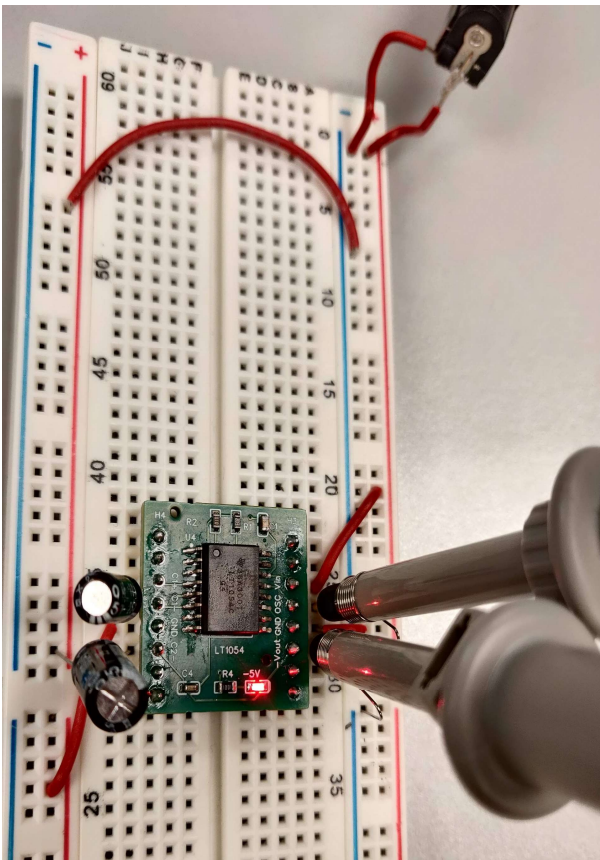
Cable ends are closer



$$V_{\text{coupled}} = (3.1 - 0.5)\% * V_{\text{in}} = 2.5\% * V_{\text{in}}$$

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Effect of Inverter(LT1054) on 5V power rail



$$V_{\text{noise \%}} = \frac{V_{\text{ripple}}}{V_{\text{in}}} * 100$$
$$= 62\text{mV}/5\text{V} * 100$$
$$= 1.2\%$$

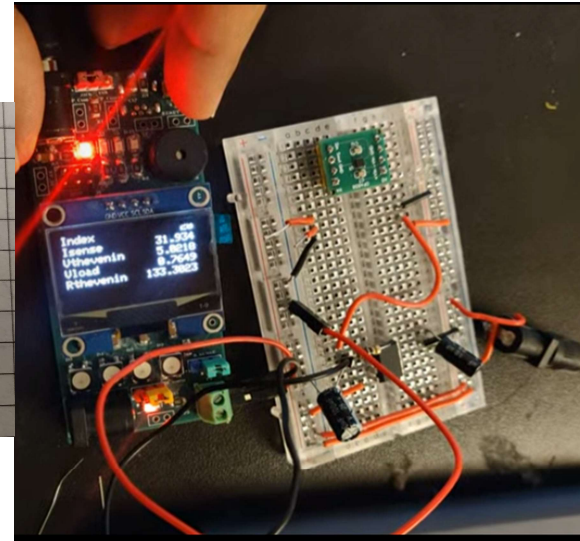
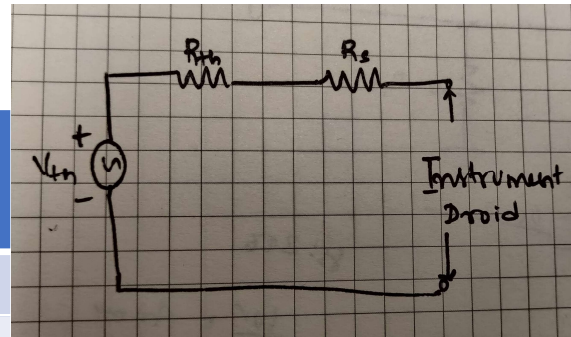
ICL7660S characteristics

Instrument droid:

Test condition: $V_s = 5V$, $C_1 = 10\mu F$ and $C_2 = 100\mu F$, $R_s = 10\ \Omega$

$R_L = 10\ \Omega$ (Instrument droid setting)

Index	$I_{\text{sense}}(\text{mA})$	$V_{\text{th}}(\text{V})$	$V_l(\text{V})$	$R_{\text{th}}(\Omega) + R_s(\Omega)$	$R_{\text{th}}(\Omega)$
1	13.066	5.022	3.2	133.2	123.2
2	26.261	5.019	1.63	120	110
20	31.964	5.021	0.76	133.3	123.3



Manual:

Test condition: $V_s = 5V$, $C_1 = 10\mu F$ and $C_2 = 100\mu F$, $R_L = 100\ \Omega$, $V_{\text{th}} = -5.12V$, $V_l = -2.28V$, $R_{\text{th}} = 124.56\ \Omega$

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With 10% voltage drop: Test condition: $V_s = 5V$, $I_s = 2A$, $R_L = \sim 1190\Omega$, $I_L = \sim 2-3mA$

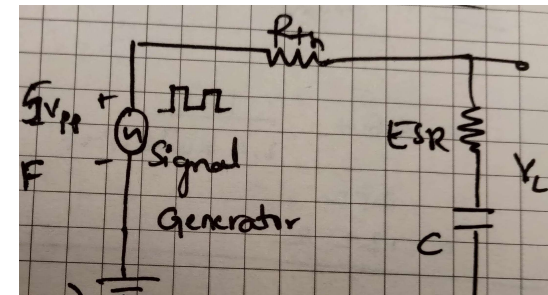
$C_1, C_2(\mu F)$	$V_{th}(V)$	$V_L(V)$	$R_L(\Omega)$	$R_{th}(\Omega)$
10	-5.2	-4.63	1220	122.9
100	-5.2	-4.77	1220	109.9
330	-5.2	-4.63	1220	141

From data sheet ($V_s = 5V$, V_L will be $-3.5V$ which is not 10% of V_s):

Output Source Resistance	R_{OUT}	$I_{OUT} = 20mA$	-	60	100	Ω
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Here $R_0 = R_{th}$

$$R_0 \cong 2 \times R_{SW} + \frac{1}{f_{PUMP} \times C_1} + 4 \times ESR_{C1} + ESR_{C2}$$



$C_1, C_2(\mu F)$	$f_{PUMP}(KHz)$	$ESR(\Omega)$	$R_0(\Omega)$	Similarity wrt load
10	5	5	111	No load
	4	7	131	With load
100	5	7.7	86	No load
	4	10	101	With load

After comparing the response of inverter with capacitor ESR, can say that the Thevenin resistance depends on pump capacitor and frequency of switching those capacitors. Ultimately R_{th} (test circuit) is around 120-130 Ω at 10 μF pump capacitors

Noise due to oscillation

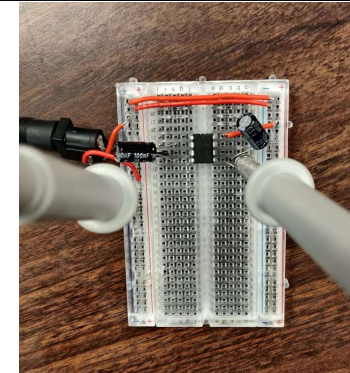


Test condition: $V_s = 5V$, $C_1 = 10\mu F$ and $C_2 = 10\mu F$

$R_{th} = 137.3\Omega$

Load	Oscillator frequency	Vout	Noise in power rail	Noise in inverted output
open	~5KHz	-5.23V	554mV	25.2mV
220 Ω	~4.38KHz	-3.22V	300mV	220mV

$$V_{\text{fluctuation}\%} = \frac{V_{\text{ripple}}}{V_i} * 100 = \frac{220\text{mV}}{|-3.22\text{V}|} * 100 = 6.8\%$$



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Test condition: $V_s = 5V$, $C_1 = 100\mu F$ and $C_2 = 100\mu F$

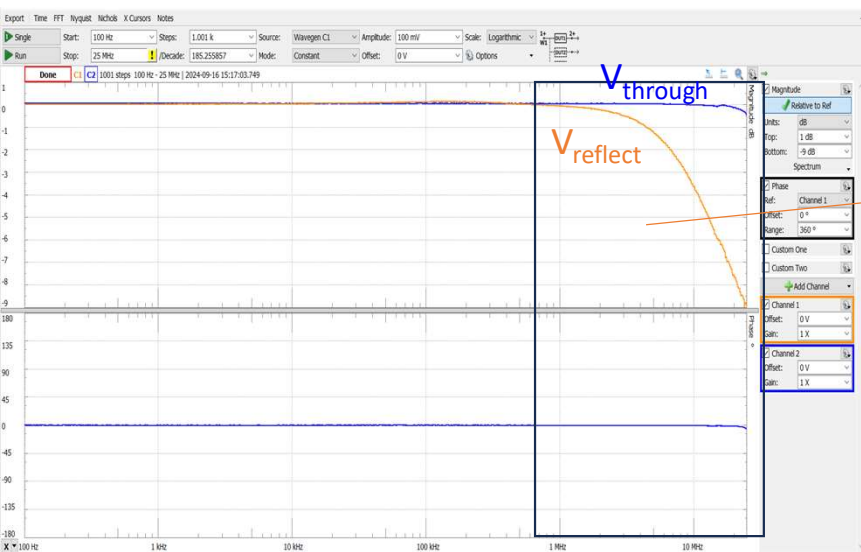
Load	Oscillator frequency	Vout	Noise in power rail	Noise in inverted output
open	~5KHz	-5.29V	129mV	227mV
220 Ω	~4.32KHz	-3.05V	81.3mV	~40mV

$$V_{\text{fluctuation \%}} = \frac{V_{\text{ripple}}}{V_i} * 100 = 40\text{mV} / |-3.05|V * 100 = 1.3\%$$

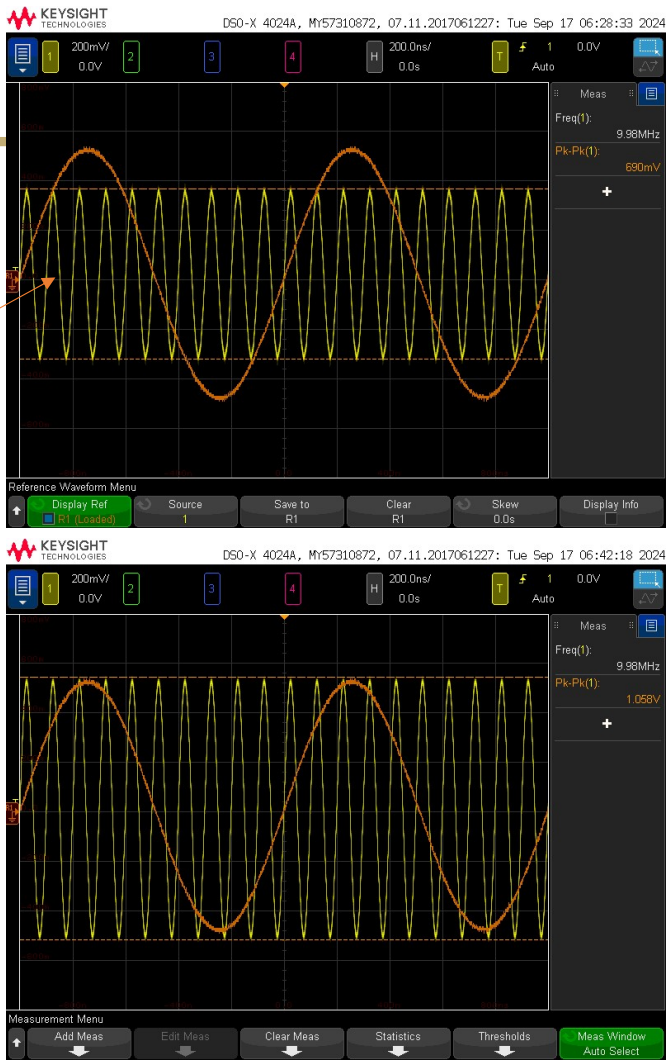
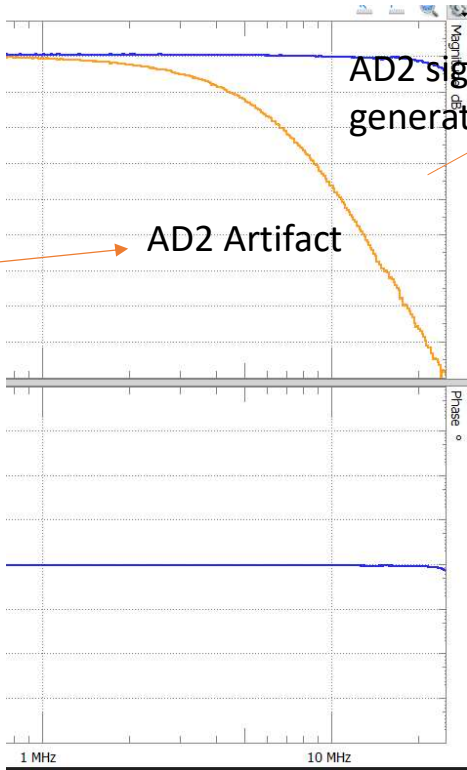
- Adding larger capacitor, decreases the oscillation and decrease in output voltage. Apparently decreased the noise in voltage rail and output.

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Bode plot at lower frequencies



The plot shows that the op-amp unity gain is dropping from 100Hz up to 25MHz.

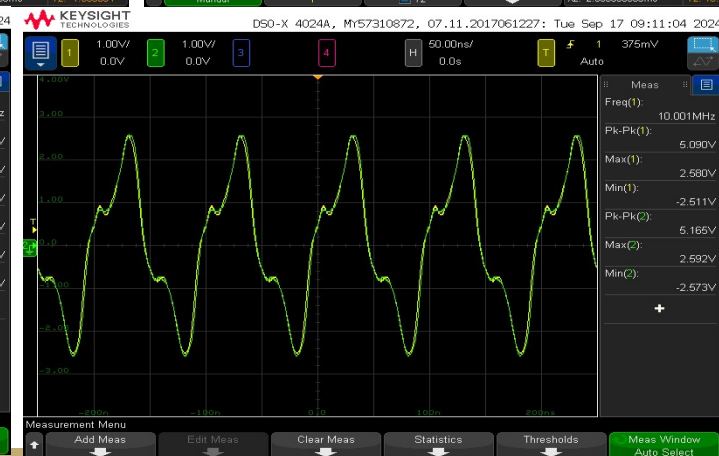
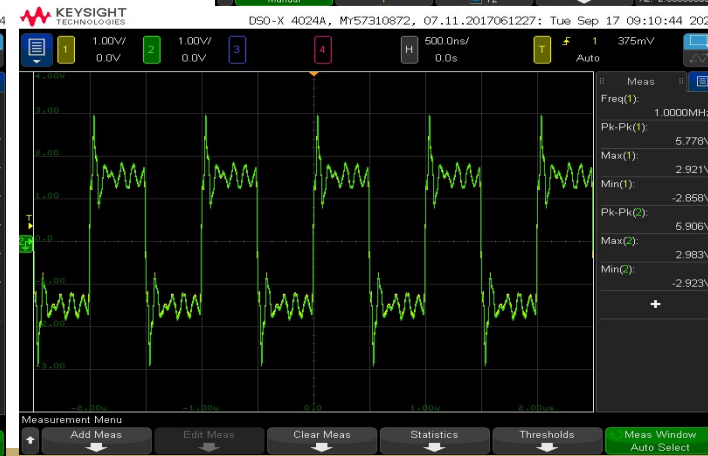
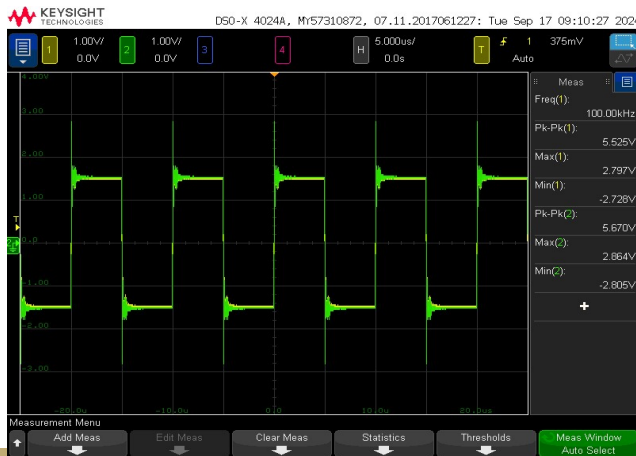
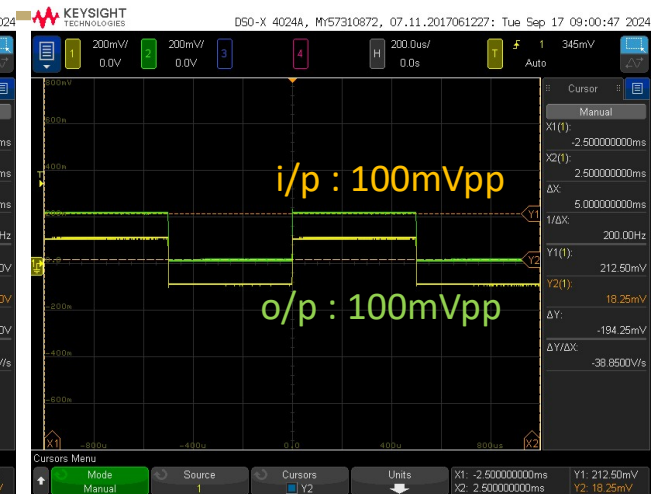
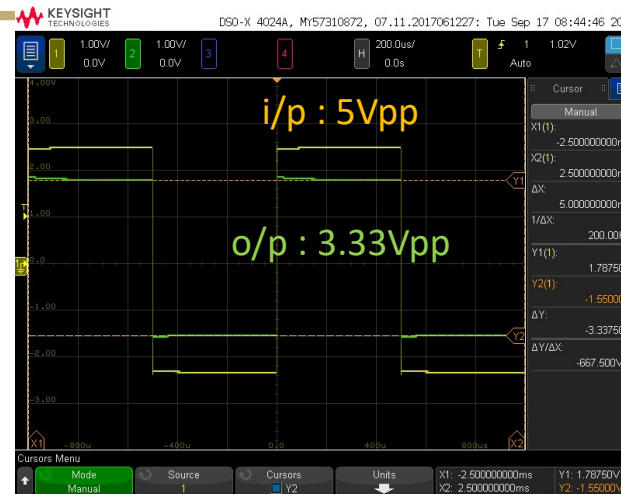


RIGOL DG822

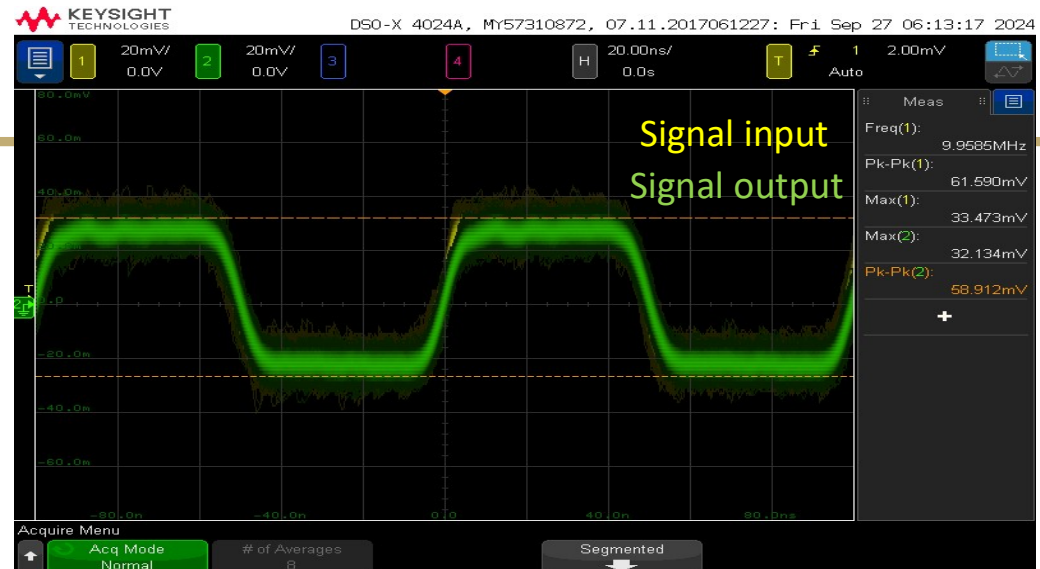
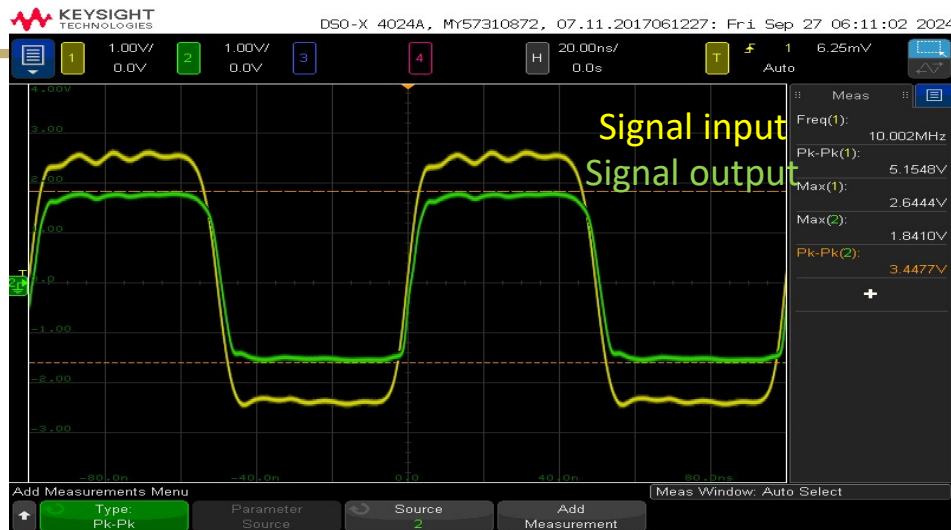
signal generator

Max output range of OPA858

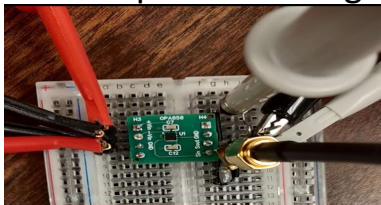
- At 5Vpp supply, measured opamp maximum peak to peak output voltage of **3.33V**, minimum of **100mV** and offset of **100mV**.
- At higher frequencies, observed high distortion in the output and losing its shape(may be due to missing harmonics -> slew rate(1ns) or bandwidth(600MHz), **need to check with sine wave input**).



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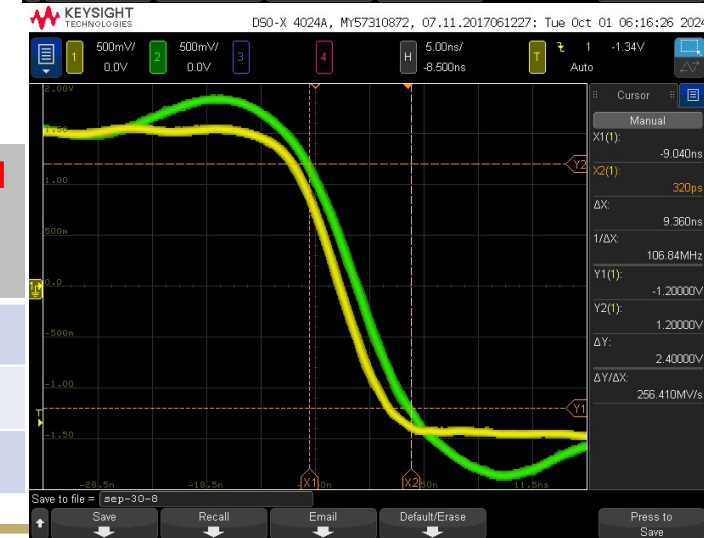
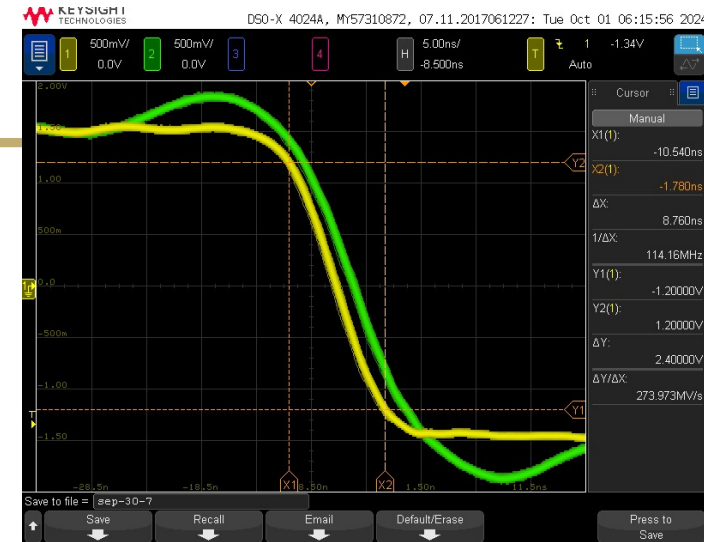
With reduced loop inductance noise(used SMA connector), at 10MHz square wave signal:



Input Vpp(V)	Output Vpp(V)
5	3.4 ± .100mV
50m	50 ± 10mV
20m	20 ± 4mV

- No DC offset seen
- Rise/Fall time degradation wrt to input signal can be seen in MHz range(Need to measure it in GHz range)

Rise/Fall time degradation

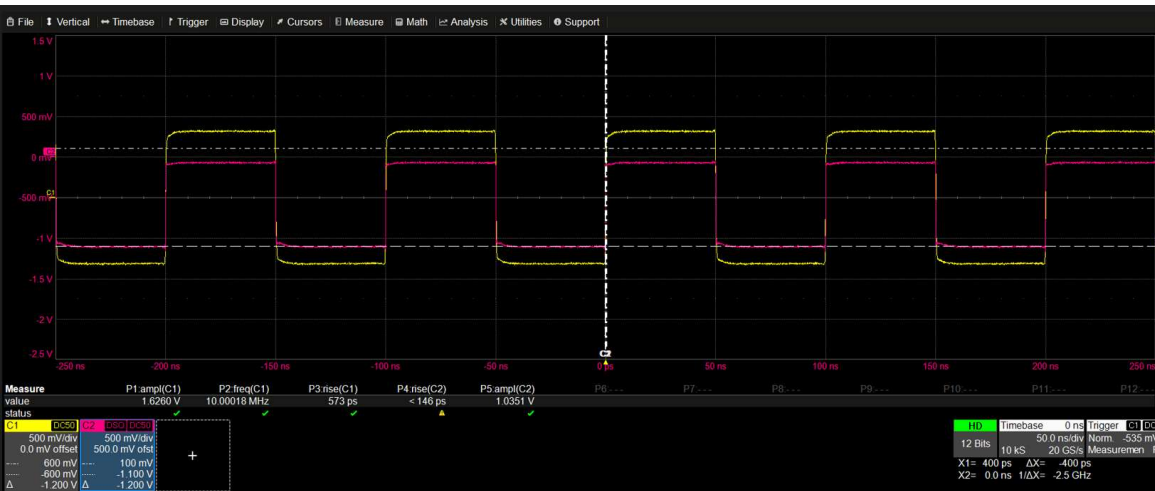


Signal in Vpp (V)	Rise time I/P (nsec)	Fall time I/P (nsec)	Rise time O/P (nsec)	Fall time O/P (nsec)	Diff in Rise time (nsec)	Diff in Fall time (nsec)
3	8.12	8.76	9.62	9.38	+1.5	+0.62
1	8.56	9.32	9.06	9.02	+0.5	-0.3
100m	8.44	9.6	9.22	8.82	+0.78	-0.78

SR	Slew rate (10% - 90%)	$V_{OUT} = 2\text{-V step}$	$\sim 2\text{V/ns}$	2000	V/ μs
t_r	Rise time	$V_{OUT} = 100\text{-mV step}$		0.3	ns
t_f	Fall time	$V_{OUT} = 100\text{-mV step}$		0.3	ns

High Speed Digital Engineering Group

Contd- With short risetime pulse generator

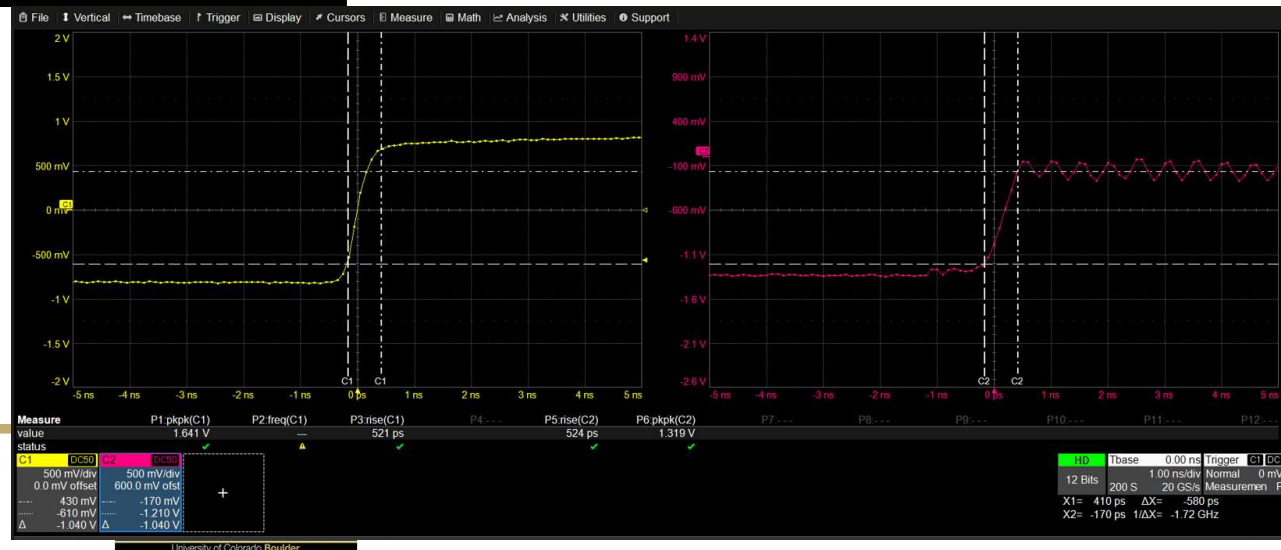


Trig out = 1.5Vpp

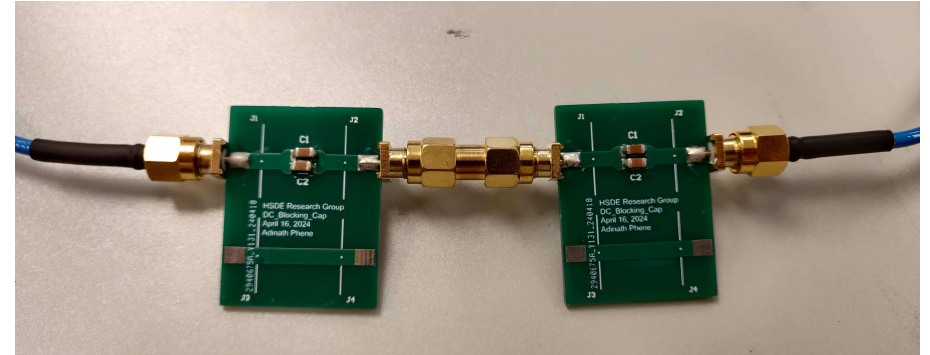
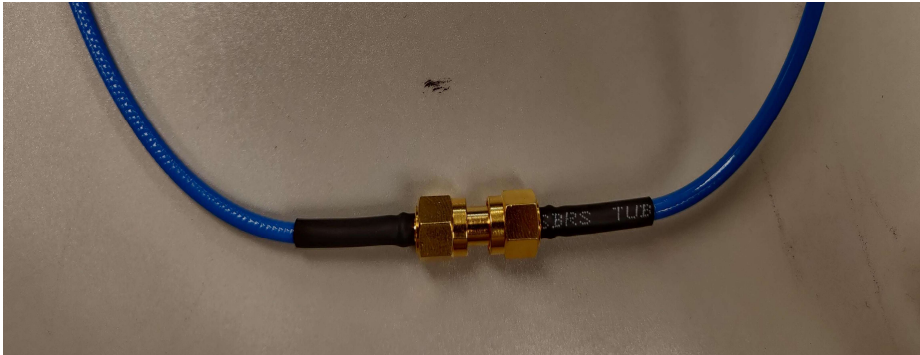
Pulse out = 1Vpp

Input signal rise time < 135psec(30psec as we datasheet)

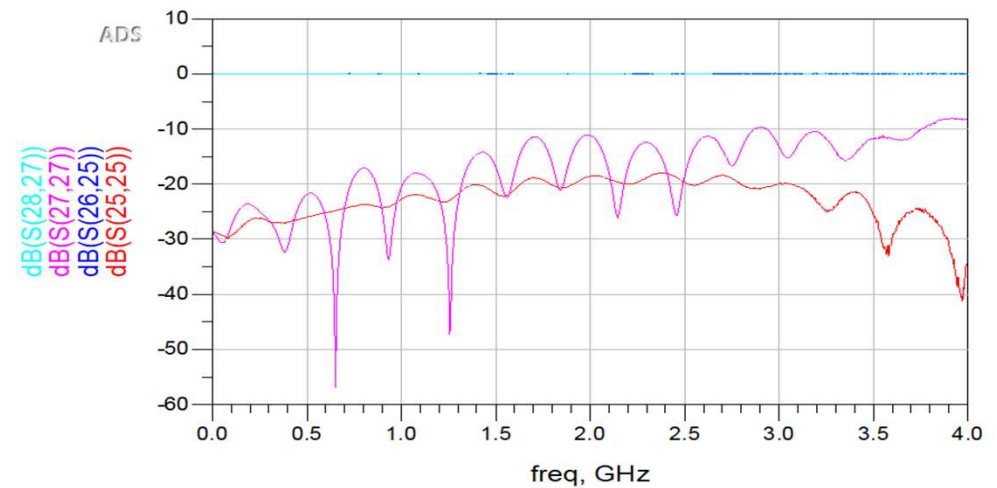
Output signal rise time ~550±30psec



Impact of DC blocking capacitor on Calibration



- More return loss($\sim 12\text{dB}$)
- More discontinuity(Magnitude of Ripples increasing with frequency)
- Yet to measure and analyze the TDR response.....



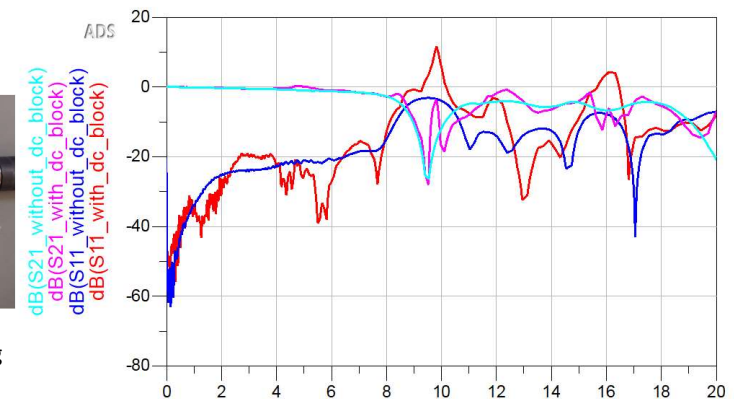
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Measured with ZNL VNA and here is the observation:

- The transmission line have a resonant structure which resonates around **9.5GHz**(0.310 inch from thermal relief)
- Launch discontinuity(capacitive impedance) is more with dc blocking structure

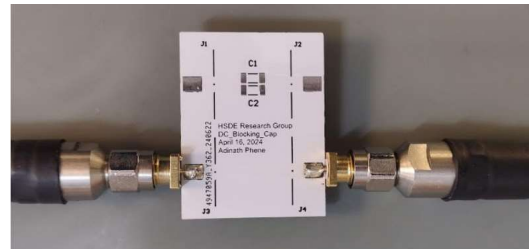


Calibrated and measured transmission line with DC blocking structure on both ends

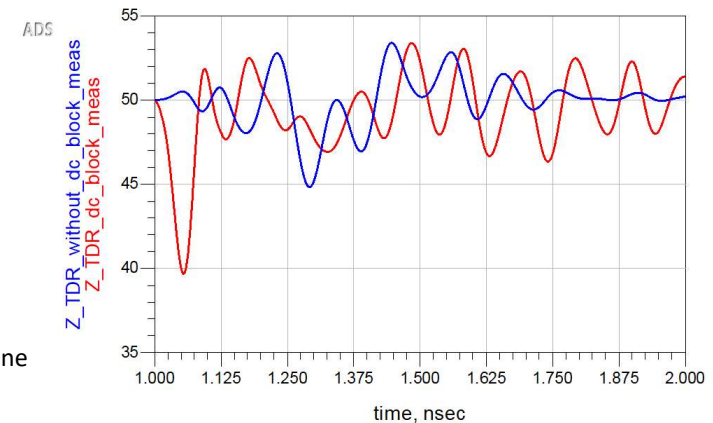


Takeaways:

- Designed active probe operates within 5GHz range(typically 2GHz)
- The insertion loss seems to 0dB over 5GHz range
- The return loss is low negative dB(Max -20dB) over 5GHz range
- No impact from the DC blocking caps to active probe measurements

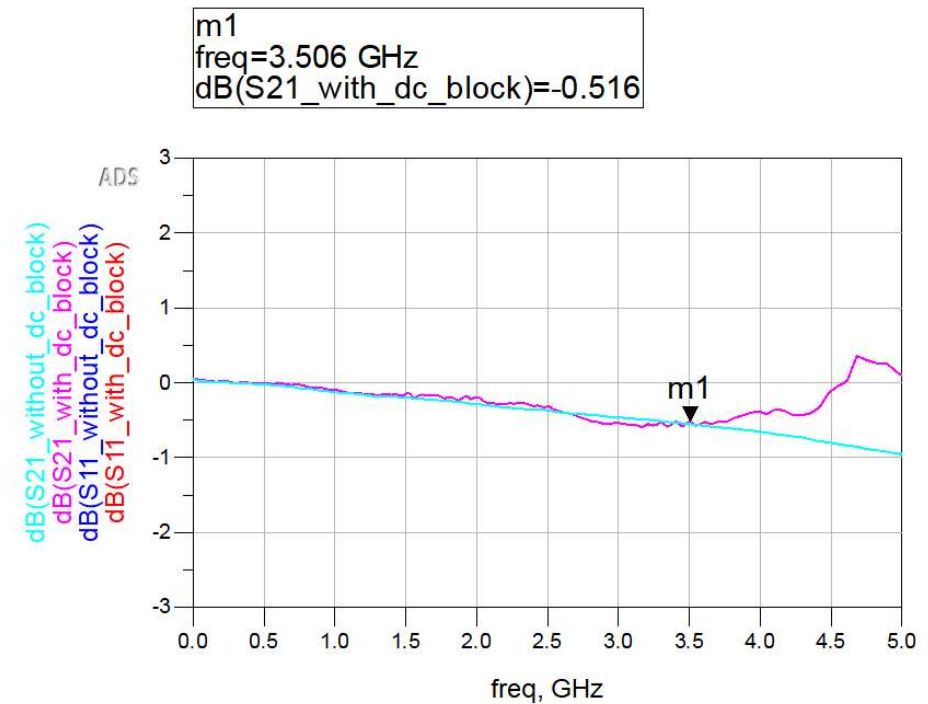


Calibrated and measured with just transmission line



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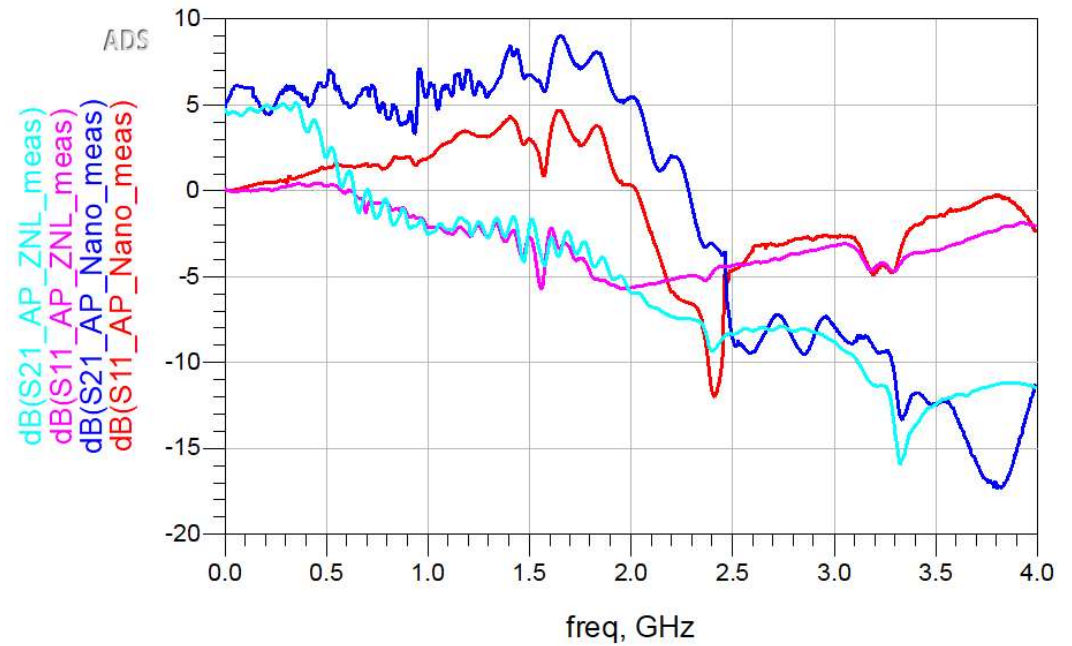
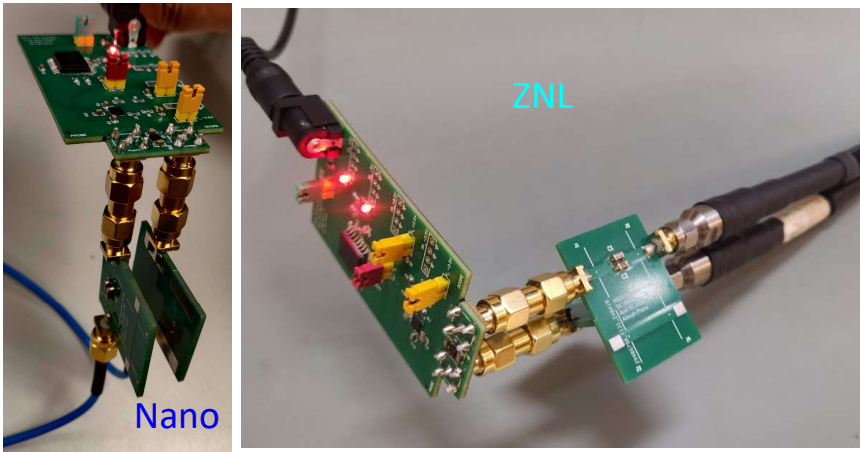
- Area of interest is upto 5GHz, within that range maximum loss observed due to dc-block is at 3.5GHz, about roughly -0.5dB
- -0.5dB = 94.4%, means nearly 6% of the input signal might get attenuated



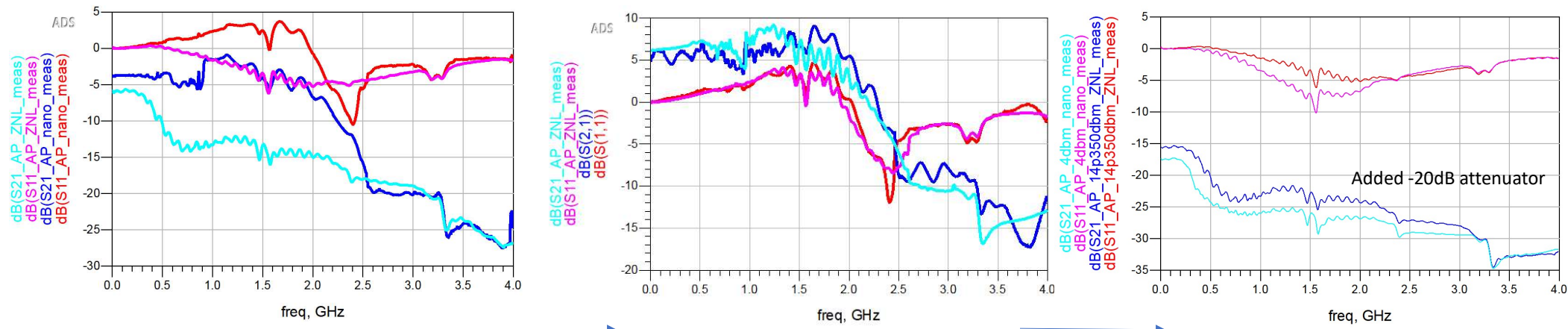
Why this behavior

Configuration on both VNAs:

- Start frequency : 3.9MHz
- Stop frequency : 3989.7MHz
- Calibration : Included dc blocking structure
- Power: 0dBm
- Sweep type: linear



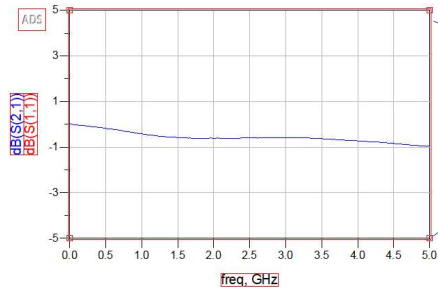
Cont



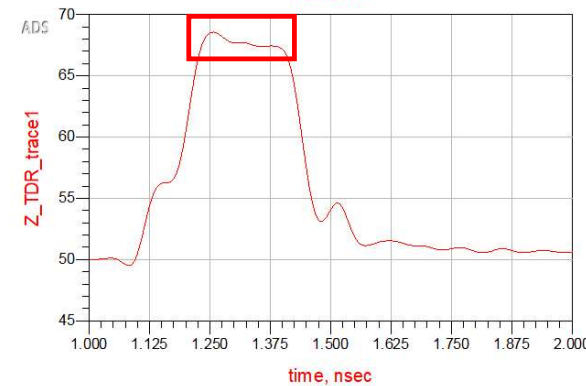
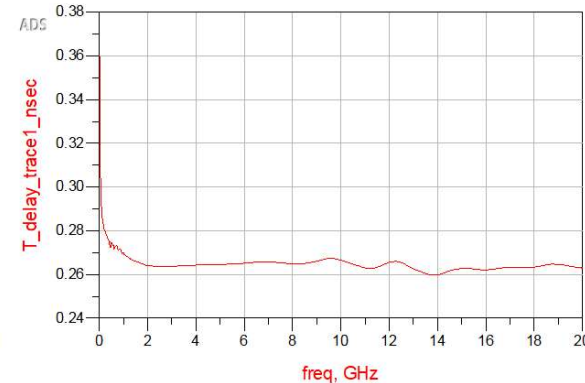
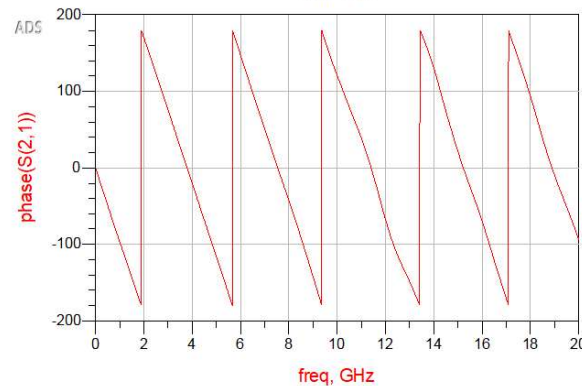
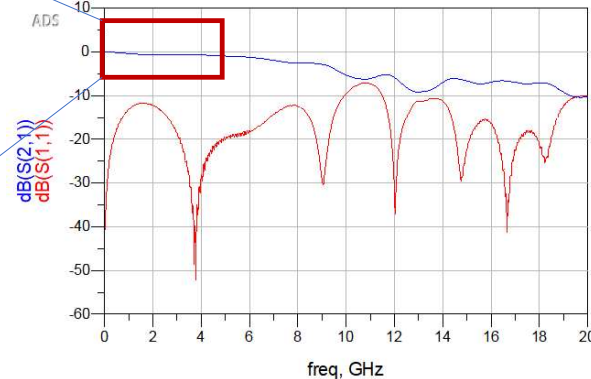
- Added 10dB attenuator and removed dc-blocking caps
- Not much changes apart from S21 attenuated by 10dB as expected
- With -10dBm (small signal) power output ($V_{out} = 200\text{mVpp}$) at ZNL VNA, able to match the Nano VNA response
- Further analysis with large signal such as 14.35dBm ($V_{out} = 3.3\text{Vpp}$) and 4dBm ($V_{out} = 1\text{Vpp}$) narrows down the operating bandwidth to 300MHz

*Note: Nano VNA signal power output was -10dBm, which is not considered in earlier analysis

Sniff board thru analysis



Observed $\sim 1\text{dB}$ attenuation over the USB3.0 operating frequency range. So, when measured with active probe rev 1, expected to see $\sim 30\%$ loss in signal strength



Lack of Spatial awareness: Instead of considering this trace as co-planar single ended (13.48mil), considered width as co-planar differential pair with 100ohm (8.02mil) resulting in $67 \pm 1\text{ohm}$ impedance

Characterization of opamp(opa858)

Step	Steps		Specified	Verified	Remark
1	Operating Voltage and Max bipolar output voltage		5.5Vpp(+2.5V to -2.5V), V_{S+} -0.9V, V_{S-} +0.9V	3.2V(+1.6V to -1.6V)	
2	Input impedance		1G(CMIR)	Very high	As per S-parameters S11 analysis
3	Input capacitance		0.6pF	4pF	As per S-parameters analysis
4	Offset		$\pm 0.8\text{mV}$	0V	Not seen any offset with 10mV resolution scope
5	Frequency	Small signal BW	1.2GHz	2GHz	As per S-parameters analysis
		Large signal BW	600MHz	300MHz	
6	Rise time		100mV/0.3ns	-	
	Slew rate		1V/0.5ns	1V/0.55ns	10% larger rise time

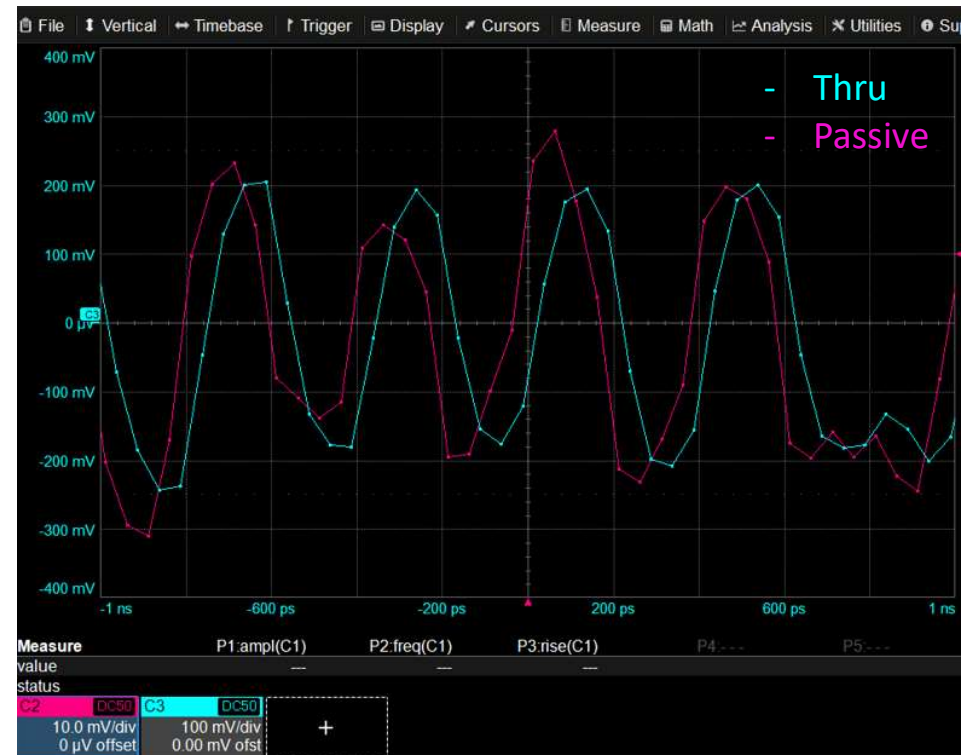
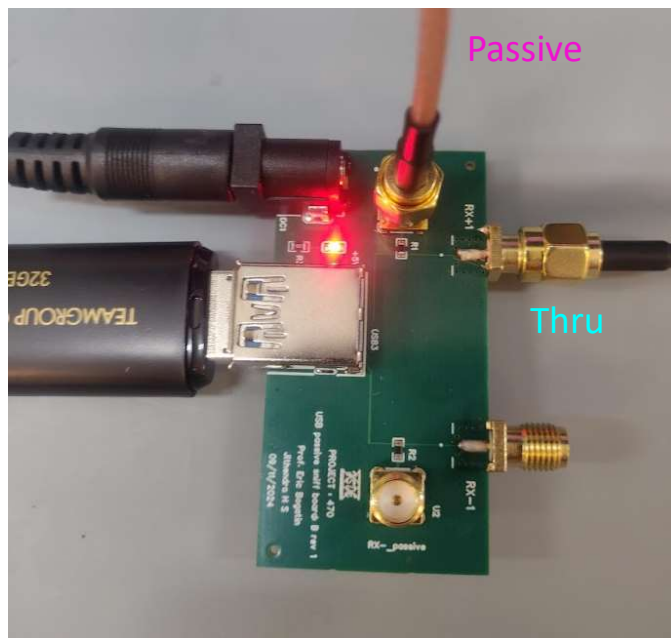
31

Opamps

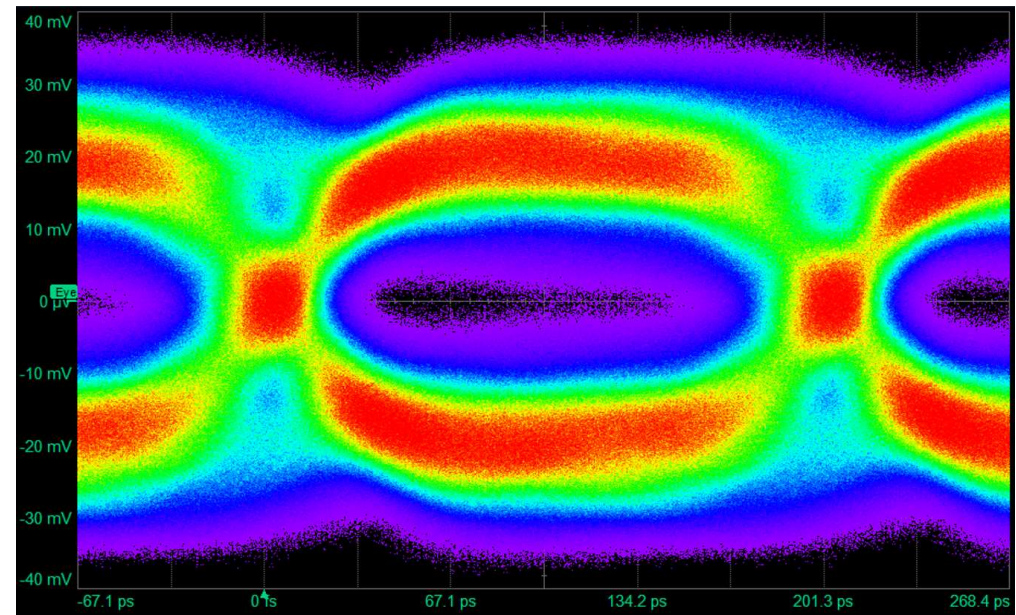
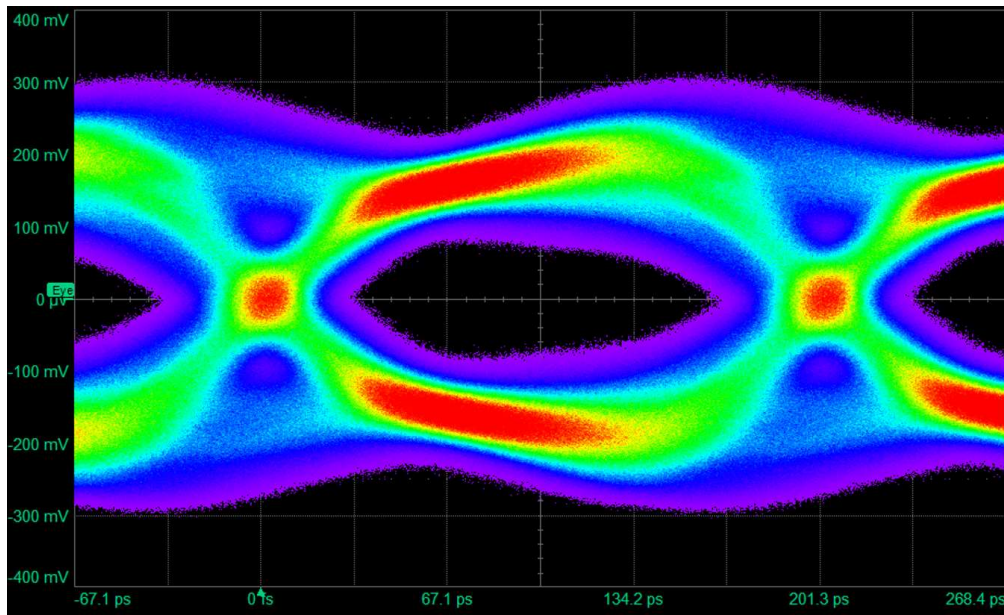
			OPA858(CMOS)	BUF802(JFET)	OPA855(BJT)
Step	Steps		Specified	Specified	Specified
1	Operating Voltage and Max bipolar output voltage		5.5Vpp(+2.5V to -2.5V), V _{s+} -0.9V, V _{s-} +0.9V	13Vpp(-6.5V to +6.5V)	5.25Vpp
2	Input impedance		1GΩ (CMIR)	50 GΩ(@100MHz)	2.3MΩ
3	Input capacitance		0.6pF	2.4pF(@100MHz)	0.6pF
4	Offset		±0.8mV	-600mV	±0.2V
5	Frequency	Small signal BW	1.2GHz	3.1GHz	2.5GHz
		Large signal BW	600MHz	3.1GHZ(upto 2Vpp)	850MHz
6	Rise time		100mV/0.3ns	250mV/0.15ns	100mV/0.17 ns
	Slew rate		1V/0.5ns	1V/0.14ns	1V/0.36ns
7	Input bias current(max)		5pA	25pA	18uA

Thru sniff VS Passive sniff

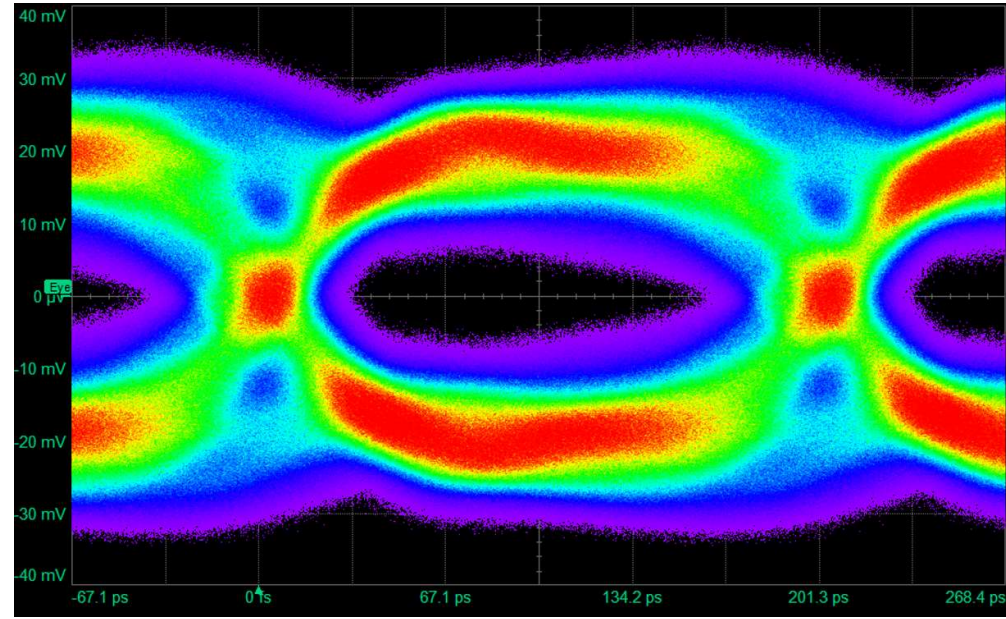
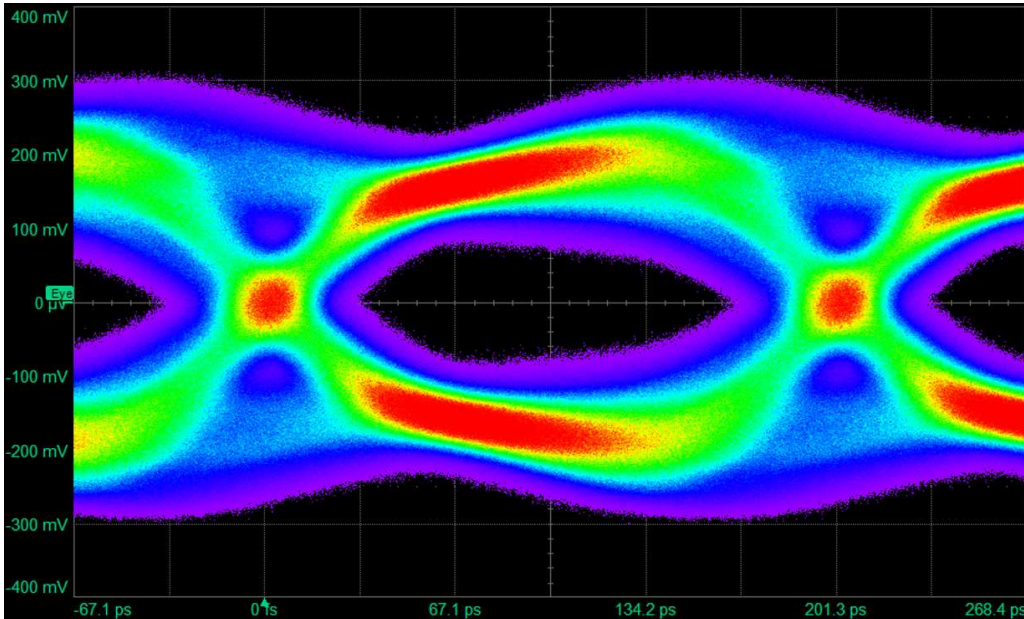
- Passive signal output follows same bit pattern of Thru, but difference is amplitude (1/10 times less) due to 450ohm voltage divider



Contd- both connected

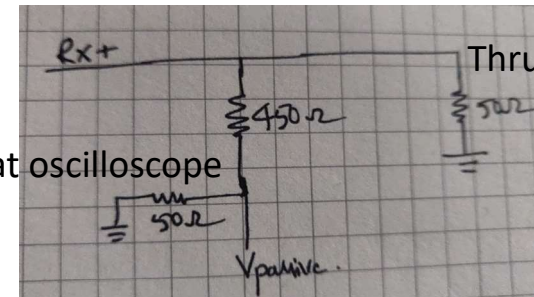


Contd- terminated by 50ohm



Passive output improved with 50ohm termination, why?

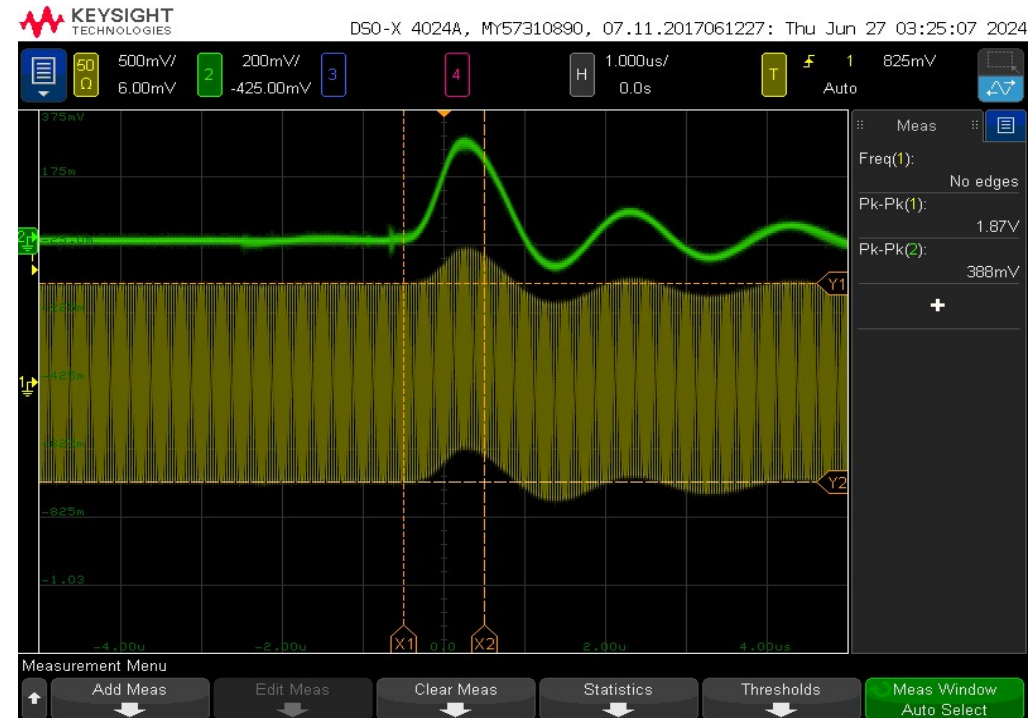
50 ohm at oscilloscope



Thru terminated

Crappy power supply

- When trying to characterize the few breakout board with one of the 5V adapter observed distortion about **388mV** at 60Hz frequency.



VNA signal using oscilloscope

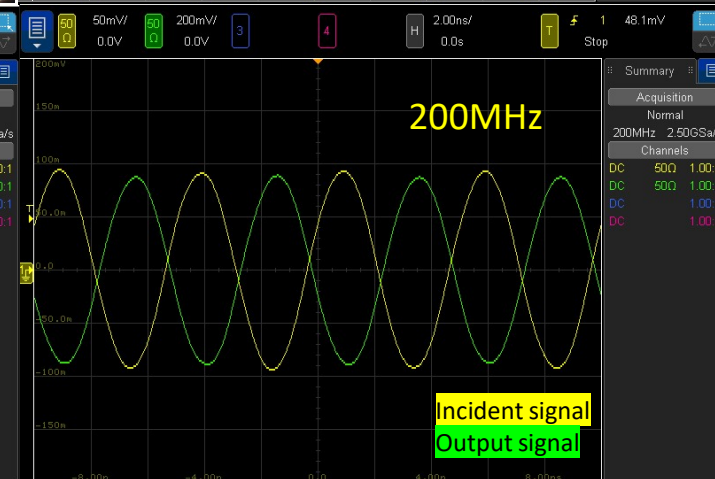
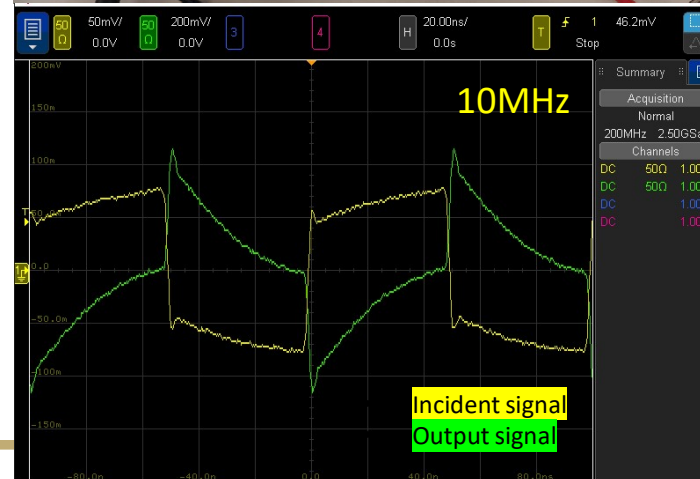
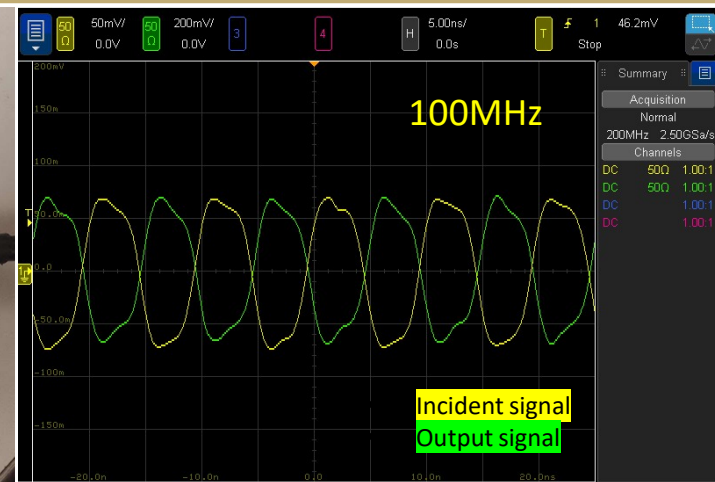
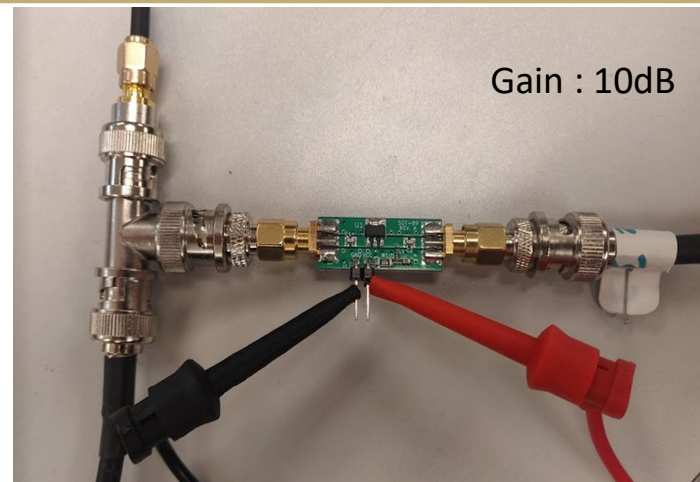
At 10MHz:

- First order square wave seen
- Low pass filter characteristics
- Pole frequency(-3dB)= $0.16/\text{time}$ at 66% of the signal, which is $\sim 10\text{MHz}$

At 100MHz and 200MHz:

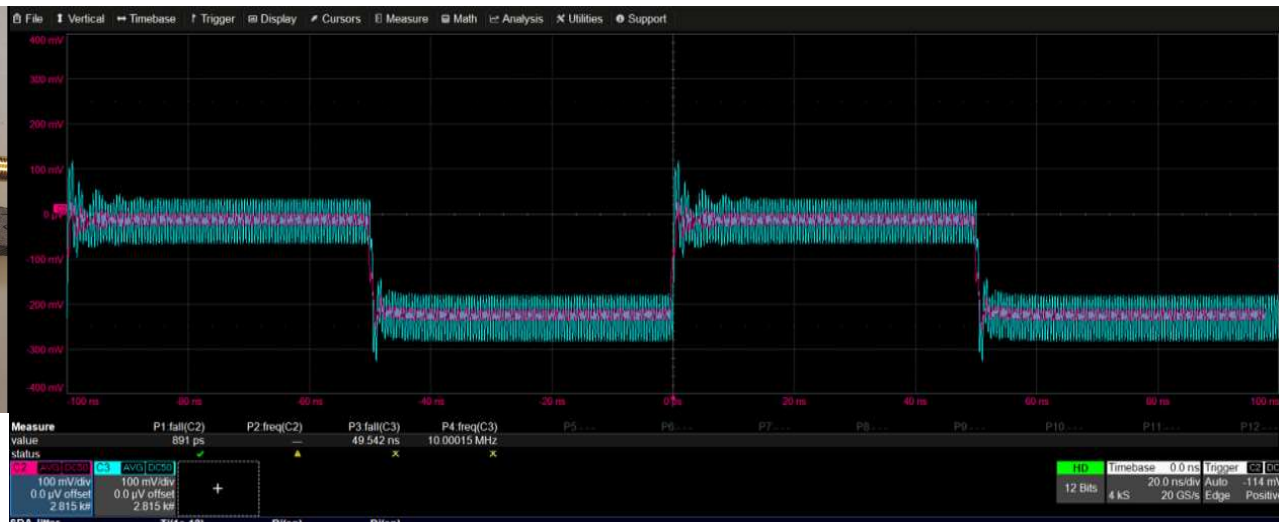
- Sine wave
- Higher frequency harmonics are filtered out

But VNA has high-pass filter that will get consider all the harmonics



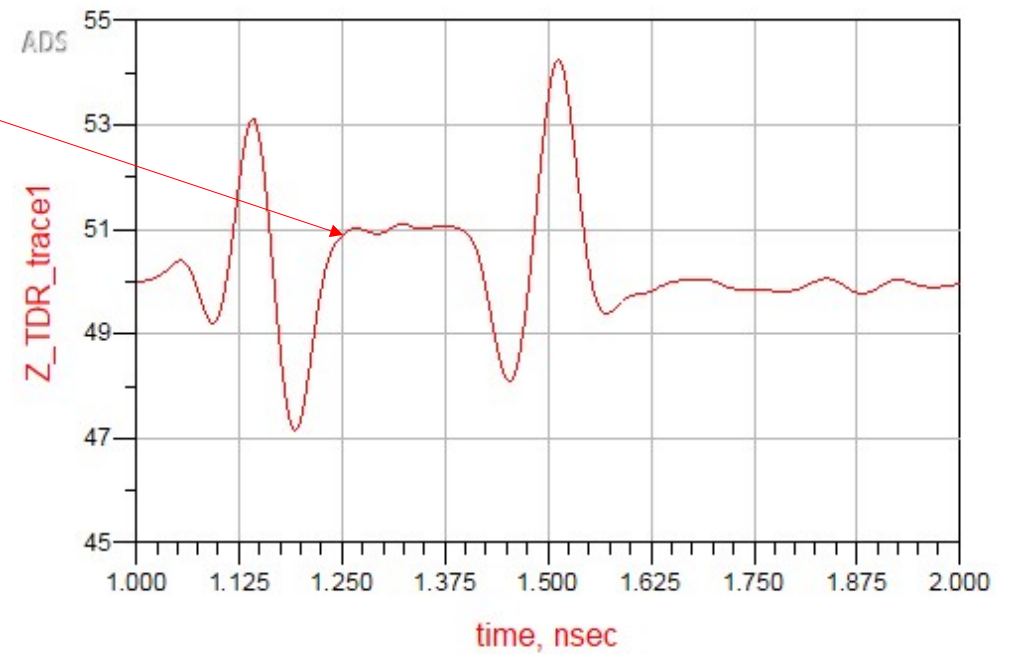
Doubt

- With small signal, not able to see twice the amplitude of incident signal?

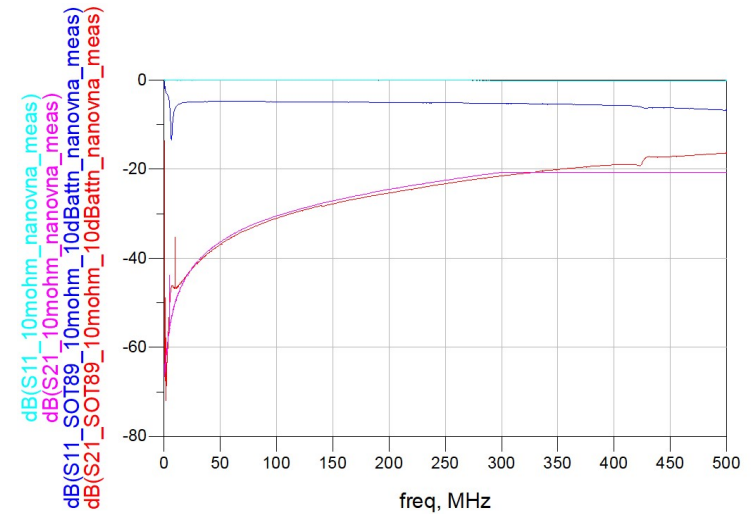
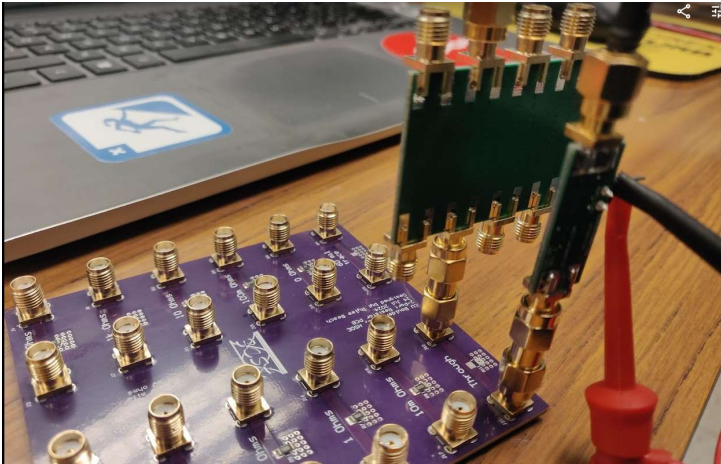


Trace impedance corrected to 50ohm

- Latest TDR output shows 51ohm
- Yet to find the attenuation of the trace

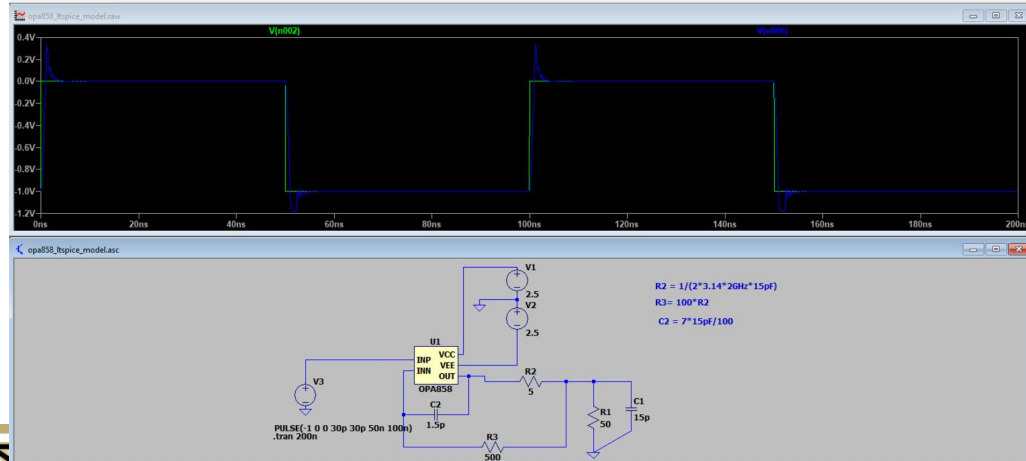
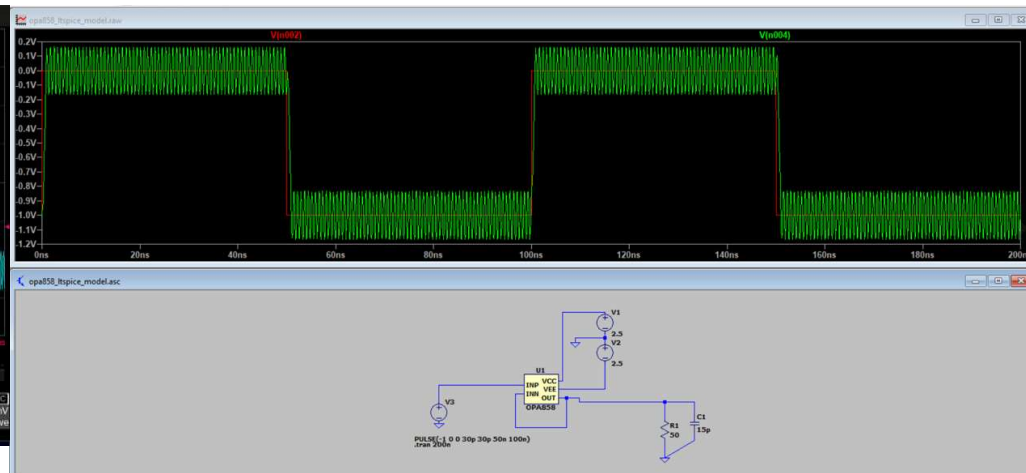
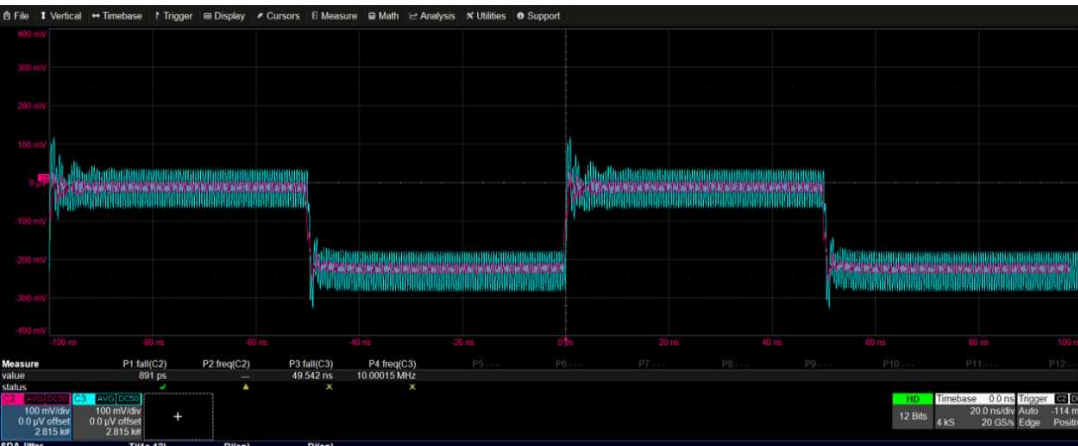


Experiment on amplifying the VNA signal



- S21 is varying ~5dB at higher frequencies
- S11 varied a lot due to the reflected signal necessity to go through op-amp output to input/feedback circuit and then reach the VNA port

Root cause and reduce oscillations



- <https://resources.pcb.cadence.com/blog/why-does-my-op-amp-oscillate>
- <https://www.analog.com/en/resources/analog-dialogue/articles/techniques-to-avoid-instability-capacitive-loading.html>
- <https://resources.altium.com/p/oscillations-clipping-and-ringing-amplifier-stability-analysis>
- <https://www.ti.com/document-viewer/lit/html/SSZTCM1>
- Analog design journal
- https://e2e.ti.com/blogs_/archives/b/thesignal#pifragment-213697609-a
- <https://www.ti.com/video/4080288067001>



High Speed Digital Engineering Group

Questions?